

CORAL: A Framework for Structuring Individual Cognitive Processes as AI Reasoning Processes

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Current human-AI collaboration faces a structural constraint in which users' thinking processes are not adequately reflected in AI reasoning processes. This study proposes an approach to structure and verbalize individual thinking processes and implement them as AI reasoning processes, along with the CORAL framework that realizes this approach. As a proof of concept, we implemented the S-OPE configuration using CORAL and conducted a comparative evaluation study across four conditions. On complex problem-solving tasks simulating real-world scenarios (35-point rubric), the S-OPE configuration achieved statistically significantly higher scores compared to custom instructions combining multiple existing prompt engineering techniques (median 30.7 vs 25.0, effect size $r = 0.83$, $p < 0.001$). Additionally, ablation analysis confirmed an integration effect exceeding the sum of effects of individual CORAL framework components. This study demonstrates the feasibility of structuring and implementing individual thinking processes in natural language.

Introduction

Current Limitations in Human-AI Collaboration

Current mainstream forms of human-AI collaboration face a problem: users' domain expertise and thinking processes are not adequately reflected in AI reasoning processes. This constraint is a structural issue arising from the fact that existing AI utilization approaches are external tools oriented toward general-purpose functionality. Prompt engineering and existing agent systems have achieved significant results in efficient task support; however, no mechanism exists to integrate individual users' cognitive characteristics or problem-solving styles into AI systems. Consequently, users remain in the position of "operators," and collaboration with AI is limited to standardized interactions.

This challenge is significant from the perspective of human-centered design principles in the HCI field. Flavell (Flavell 1979) defined metacognition as “thinking about thinking,” articulating the importance of the ability to monitor and control cognitive processes. Zimmerman’s (Zimmerman 2002) self-regulated learning theory systematized the process of acquiring expertise through self-reflection. These theories provide a foundation for reflecting individual cognitive processes in AI systems.

An Approach: Structuring Cognitive Processes as AI Reasoning Processes

This study proposes an approach to achieve effective human-AI collaboration by structuring and verbalizing individual thinking processes and implementing them as AI agent reasoning processes. This approach enables users to explicitly describe their domain expertise, personal thinking processes, and cognitive styles in natural language and implement them as executable custom instructions for AI.

As a means to realize this approach, this study proposes the CORAL (Cell-based Organized Reasoning with Adaptive Linkage) framework. The CORAL framework enables users to implement their own thinking processes as AI agent reasoning processes using natural language, allowing users to construct personalized human-AI collaboration systems themselves. Furthermore, as a proof of concept, we present an implementation example of CORAL and quantitatively demonstrate through a statistically designed comparative study that the CORAL implementation exhibits superior performance on complex problem-solving tasks that simulate real-world scenarios.

The “complex problem-solving tasks simulating real-world scenarios” mentioned here do not disclose all information from the outset but reveal information progressively. In real-world problem-solving, it is rare for all relevant information to be available from the start; only through questioning, deepening inquiry, and forming context through dialogue does the full picture of the problem become clear. The evaluation tasks in this study incorporate an information disclosure protocol as a core design element to simulate this real-world complexity.

The approach of this study represents an integration of personalization technologies and end-user development in the HCI field, achieving personalization adapted to users’ cognitive characteristics at the higher level of human thinking processes and AI reasoning processes. Furthermore, by practicing open science and making all data and all custom instructions used in the research publicly available, we ensure transparency and verifiability, paving the way for subsequent researchers.

Main Contributions

This study contributes to the HCI field in the following four aspects:

1. **Proposal of a Structuring and Implementation Approach:** We propose an approach to structure and verbalize individual thinking processes and implement them as AI reasoning processes in the custom instruction layer.
2. **Construction of the CORAL Framework:** We propose a framework that integrates the division of custom instructions (Cell) with a context inheritance protocol (TIDE) that maintains semantic coherence after division.
3. **Realization of Natural Language End-User Development:** We provide an environment that enables domain experts to implement their own thinking processes in AI without technical expertise.
4. **Quantitative Demonstration of Effectiveness:** To evaluate frameworks at the custom instruction layer, we provide a task design example that simulates real-world problem-solving where information is progressively disclosed through dialogue, and conduct a comparative evaluation study across four conditions (N=36) to quantitatively evaluate the effectiveness of the CORAL framework.

Relevance to HCI Community

This study contributes to the HCI readership by demonstrating the feasibility of AI interactive systems that integrate users’ personal cognitive characteristics and expertise into AI systems and adapt to user-specific thinking processes.

Related Work

Technical Foundations and Framework Evolution

In maximizing LLM capabilities, reasoning frameworks such as Chain-of-Thought (Wei et al. 2022), Tree of Thoughts (Yao, Yu, et al. 2023), and ReAct (Yao, Zhao, et al. 2023) have demonstrated strategic thinking and action integration. These serve as technical foundations for CORAL.

Orchestration frameworks such as LangChain (“Introduction | LangChain” n.d.), AutoGen (Wu et al. 2024), and CrewAI (“Introduction” n.d.) target efficiency and general-purpose applicability as design goals and assume implementation by programmers. In contrast, the structuring and implementation of individual thinking processes that this study aims for requires different design requirements: domain experts directly implementing their thinking processes in natural language.

Technical Differences from Prompt Chaining: The CORAL framework is an extension of prompt chaining but differs technically in the following respects. First, the target of division differs. While prompt chaining divides “tasks,” CORAL divides “custom instructions (instruction text).” Second, the motivation for division differs. While prompt chaining aims for

efficient task processing, CORAL aims to avoid attention control breakdown in LLMs caused by large custom instructions. Third, the design of inheritance differs. While prompt chaining passes previous output as-is or in summarized form, CORAL inherits context semantically structured across six perspectives. These technical differences give CORAL a logical necessity of “division -> need for inheritance -> TIDE protocol.” The CORAL framework is technically a form of natural language custom instructions. However, it is simultaneously intended to bring about a qualitative transformation by giving cognitive scientific structure to custom instructions.

Current Human-AI Collaboration: Structural Limitations and Approaches

Current mainstream human-AI collaboration follows an “external tool paradigm” and contains inherent structural limitations. Tankelevitch et al. (Tankelevitch et al. 2024) demonstrated that generative AI places high metacognitive demands on users across four stages: prompt formulation, output evaluation, iterative refinement, and workflow integration. This cognitive load stems from the structure of treating AI as an external tool. Existing metacognitive support approaches contribute to reducing cognitive load, but the framework of viewing AI as “a tool to be operated from outside” remains unchanged.

Similar constraints exist in the integration of EUD and AI. Esposito et al. (Esposito et al. 2023) pointed out difficulties in technical skill acquisition, black-box nature, and lack of adaptability. Kelly (Kelly 2006) indicated that establishing evaluation axes to measure true personalization is a challenge. The cognitive characteristic-based system adaptation examined by Steichen et al. (Steichen and Fu 2019) also remains an “external adaptation” approach.

From External Tools to Structured Collaboration

In response to the challenges indicated by these prior studies—structural limitations of external tool-type collaboration, cognitive load issues, and difficulties with personalization—this study proposes the CORAL framework. In the CORAL framework, users structure and verbalize their own thinking processes and implement them as AI reasoning processes, making it possible to design and control the structure of cognitive load generation. While existing explainable AI aims to improve transparency from the system side, CORAL enables proactive understanding and control from the user side. By adopting natural language description of thinking processes, it presents a new direction distinct from programming-oriented EUD. The next chapter explains the technical details of the CORAL framework.

CORAL Framework

Framework Architecture

The CORAL framework is a mechanism that realizes multi-stage dynamic reasoning processes under a single LLM and single session, providing a structure that can flexibly adapt to users’ personal thinking characteristics and task requirements. Specifically, the CORAL framework is a composite of custom instructions called Cells that have special requirements, and stably achieves coordination between Cells through a protocol called TIDE. Cells are explained in Section , and the TIDE protocol is explained in Section . Additionally, the CORAL framework incorporates a feedback loop that supports iterative improvement by users. This point is explained in Section .

Clarification of Technical Positioning: The CORAL framework differs fundamentally in design philosophy from multi-agent systems such as LangChain and AutoGen. While multi-agent systems pursue efficiency and scalability through parallel and distributed processing by multiple LLM instances, CORAL assumes sequential processing within a single LLM and single session, with the design goal of maintaining continuity and semantic consistency in task processing. This choice is necessarily derived from the core concept of “structuring individual thinking processes and implementing them as AI reasoning processes.” Deep thinking cannot be parallelized—just as human cognitive processes unfold sequentially, the AI reasoning processes that structure and implement them must also chain sequentially.

The overall conceptual diagram of the CORAL framework is shown in Figure 1.

The Cell-based architecture divides complex tasks into manageable functional units, drawing on the separation of concerns principle in software development. Each Cell is defined as a custom instruction with a specific role, responsible for one cognitive or thinking process. CORAL is not constrained to a specific number of Cells or a fixed configuration. Users can freely add, delete, modify, and connect Cells according to project requirements and characteristics of their cognitive processes. Various forms of coordination are possible, including layered structures and circuit structures.

Examples of flexible Cell connection patterns in the CORAL framework are shown in Figure 2.

The coordination forms between Cells are not constrained to specific architectural patterns, and can flexibly adapt to specific individual thinking processes and specific tasks with dynamic flexibility. This characteristic is called “Adaptive Linkage” in this paper. The adaptive linkage of the CORAL framework is established when the LLM selects its own routing according to the TIDE protocol for human-designed reasoning processes and changes the reasoning process. Details are provided in Section . Note that while the CORAL framework is technically capable of diverse implementations including parallel processing, considering the design philosophy discussed in Section and the core concept of “structuring individual thinking processes and implementing them as AI reasoning processes,” sequential routing is assumed for Cell groups handling deep thinking and reasoning.

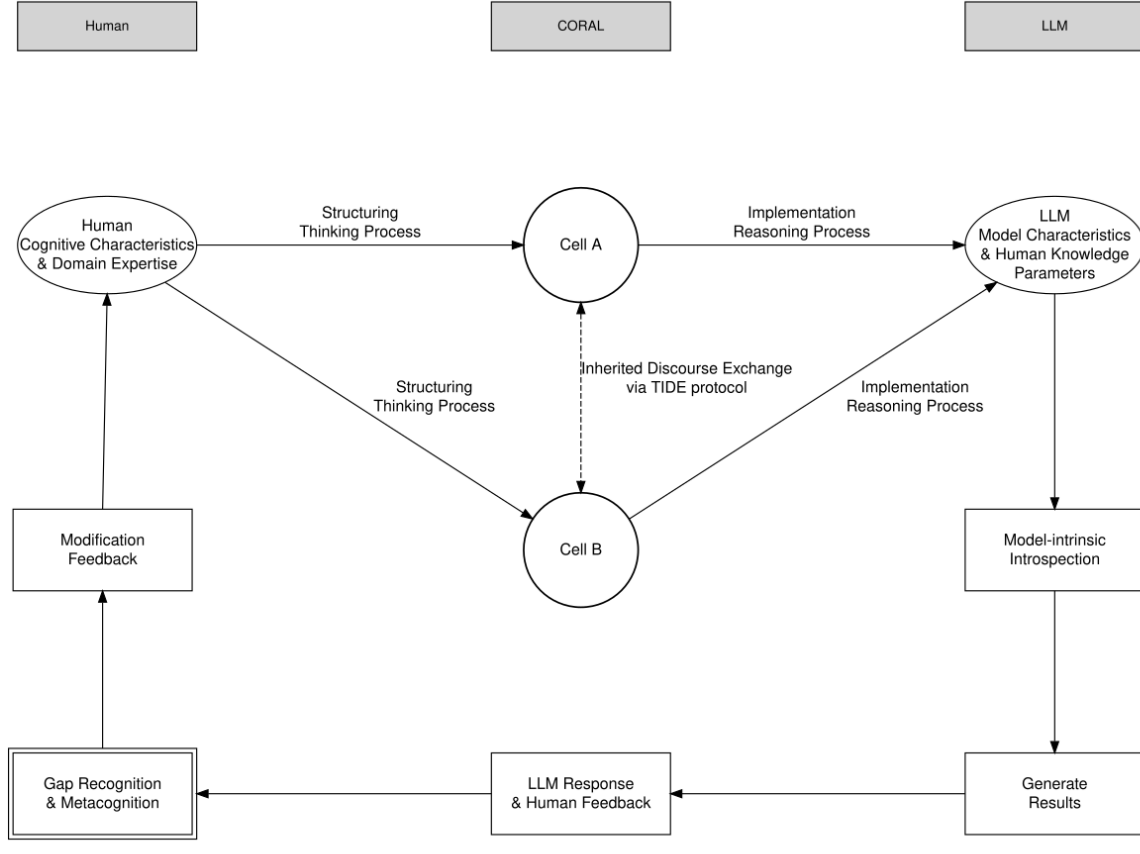


Figure 1: CORAL Framework Conceptual Diagram: This figure illustrates how human cognitive characteristics are structured as Cell-based AI reasoning processes. The left column shows human domain expertise and cognitive characteristics being structured into Cells (Cell A, Cell B) through thinking process verbalization. The middle column represents the CORAL framework’s Cell architecture with TIDE protocol enabling inherited discourse exchange between Cells. The right column shows how LLM (Large Language Model) characteristics process and generate results. The feedback loop at the bottom enables iterative refinement through gap recognition and metacognitive awareness, supporting continuous improvement of the human-AI collaboration.

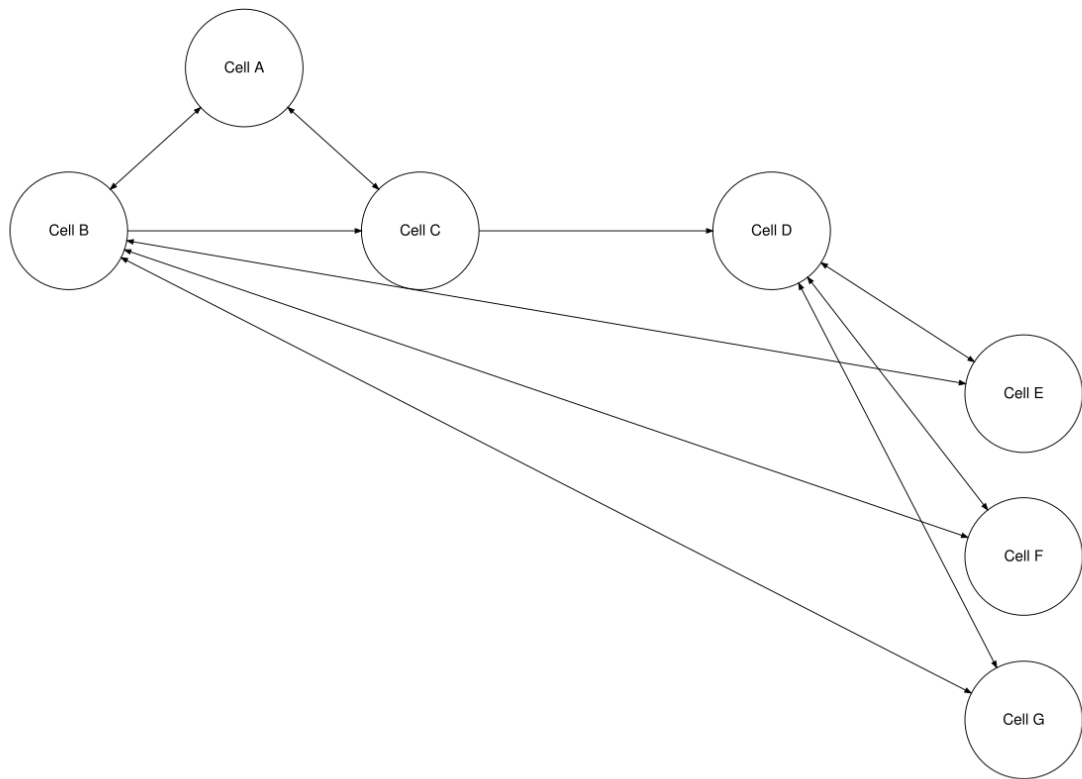


Figure 2: Flexible Cell Connection Patterns in CORAL: Examples of collaborative, hierarchical, and dual-orchestrated configurations. Cell B and Cell D serve as orchestrators, delegating tasks to other Cells based on task nature and complexity.

Cell: The Fundamental Building Block

Cell Definition and Concept

A Cell in the CORAL framework is a special custom instruction that explicitly assumes input and output. The differentiating factors from general custom instructions are two design principles: “Input-Output Oriented Design” and “Explicit Reasoning Process Definition.”

1. Input-Output Oriented Design

- Each Cell is designed with clear input and output assumptions for users or other Cells. This design philosophy enables dividing complex cognitive processes into manageable functional units and standardizing coordination between Cells. Coordination between Cells is standardized by the TIDE protocol described below, achieving consistent context inheritance and state transitions. This standardization enables multiple Cells to function as an integrated system and perform complex tasks while maintaining long-term contextual consistency. A clear information flow is designed where inputs received include inherited context from the previous Cell and user input, and outputs generated include inherited context for the next Cell and responses to the user.

2. Explicit Reasoning Process Definition

- Each Cell has its reasoning process described in natural language in the form of “what to do (Do).” This expresses individual thinking patterns and professional judgment criteria as executable logic for AI. While conventional custom instructions remain as general-purpose directives, Cells implement personalized reasoning flows that reflect users’ specific cognitive processes and problem-solving approaches.

Note that Cells in the CORAL framework are categorized into two types based on their function. **General Cells** (Orchestrator, Planner, Executor, etc.) are designed to reduce users’ cognitive load, whereas the **Sensemaker Cell** is exceptionally designed to intentionally increase cognitive load and promote metacognition (thinking about thinking). This dual strategy of cognitive load is detailed in Section .

The conceptual structure of Cells in the CORAL framework is shown in Figure 3. As illustrated, each Cell consists of three sections: the Input Section receives inherited context from the previous Cell via the TIDE protocol, the Reasoning Process Section executes the Cell’s specific cognitive or processing logic, and the Output Section generates inherited context for the next Cell. The Execution Flow arrow indicates the sequential processing direction within a Cell.

Platform Independence and Technical Implementation

Cells can be implemented on any platform that has the capability to explicitly invoke custom instructions. This versatility gives CORAL the potential to develop as a technology for

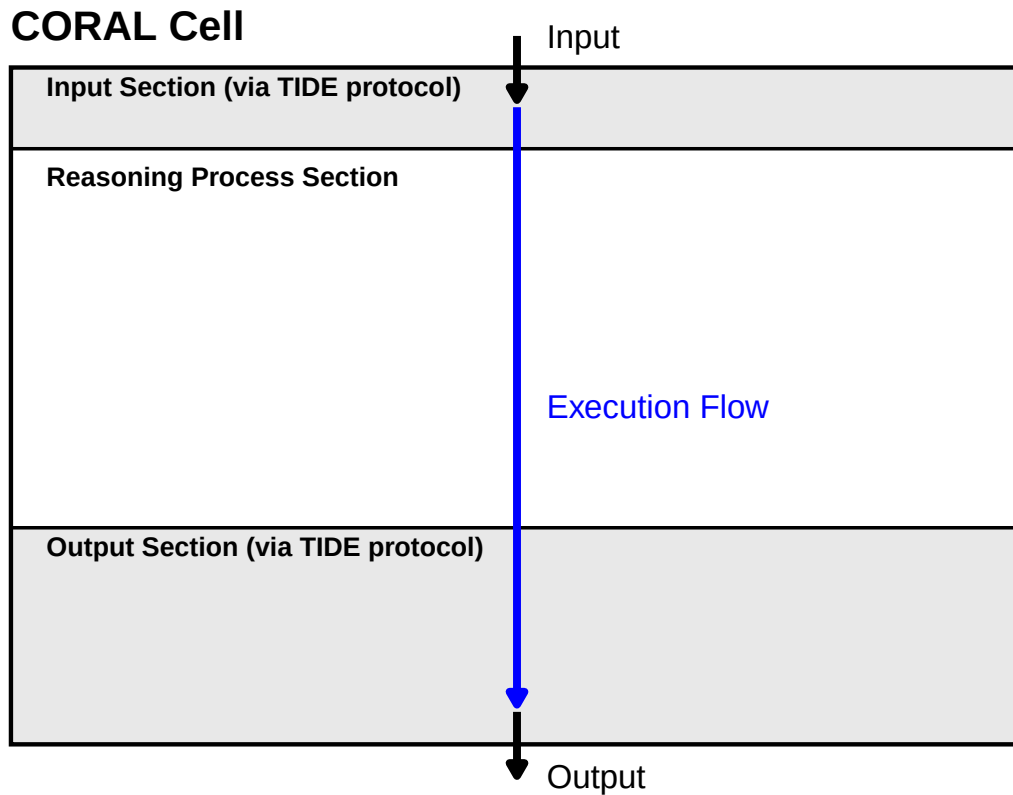


Figure 3: Conceptual Structure of a CORAL Cell: This figure illustrates the input-output oriented design, explicit reasoning process section, and TIDE protocol integration that characterize each Cell in the CORAL framework.

structuring and implementing individual thinking processes that is not dependent on specific platforms.

In this study, as an implementation example in the Cursor environment, we adopted Cell definitions in `.mdc` format and Cell invocation through the `fetch_rules` tool. Each Cell is managed as an independent file, dynamically loaded and executed as needed. This implementation approach facilitates the addition, modification, and deletion of Cells, promoting continuous improvement cycles by users.

As technical implementation details, each Cell file (`.mdc`) contains the following structured elements: (1) clear definition of the Cell’s role and responsibilities, (2) the format of information expected as input, (3) step-by-step description of the reasoning process to be executed, (4) the format of information to be generated as output, and (5) transition conditions to the next Cell and context to be inherited. This structuring enables end-user development where users design and implement AI reasoning processes in natural language.

The CORAL framework combines multiple such Cells to function as an integrated whole. While maintaining the specialization of individual Cells, it realizes the entire complex cognitive process through coordination via the TIDE protocol.

Cell Division: Practical Necessities from Implementation Experience

Cells in the CORAL framework are not a design choice pursuing conceptual elegance or theoretical beauty, but rather a necessary choice based on practical constraints that became apparent during implementation. We specifically explain why division into multiple “Cells” rather than a single custom instruction was technically unavoidable in realizing the structuring and implementation of individual thinking processes.

In the CORAL framework, the amount of description in custom instructions can become enormous in the process of expressing individual thinking processes. For example, in this study, we present the S-OPE configuration (see Section) implemented by the developer for paper writing purposes as an implementation example and quantitative evaluation target. If all Cells used in the S-OPE configuration were integrated as a single custom instruction, it would become a massive instruction text of approximately 38,000 characters (in Japanese) consisting of approximately 150 processing steps. An instruction text of this scale causes the following fundamental problems:

1. **Attention Control Breakdown:** The LLM’s attention mechanism cannot function effectively, and it cannot properly determine which part of the lengthy instruction to focus on. As a result, instruction compliance becomes unstable, making execution of expected reasoning processes difficult.
2. **Lack of Maintainability:** When modifying part of an instruction text exceeding 38,000 characters, it becomes extremely difficult to grasp the scope of impact that change has on other parts. This causes continuous system improvement to effectively cease functioning.

3. Loss of Reusability: Extracting specific functions (e.g., document creation, code generation, strategic dialogue) for reuse in other contexts or utilizing them in different combinations becomes practically infeasible. This significantly limits adaptability to the diversity and evolution of individual thinking processes.

Therefore, Cell division is a necessary design choice based on human cognitive limitations (difficulty in design and maintenance) and technical constraints of LLM systems (limitations in attention control). Furthermore, to make divided Cells operate cooperatively while maintaining semantic coherence, it was necessary to develop the TIDE protocol for managing information inheritance and state transitions between Cells and incorporate it as a structure into Cells, as detailed in Section .

TIDE Protocol for Context Inheritance

The TIDE (Transition with Inherited Discourse Exchange) protocol is a mechanism that achieves stable rule switching and context inheritance based on natural language, described in six steps.

Details of the six steps of the TIDE protocol are shown in Table 1.

In Step 3 of the TIDE protocol, “Inherited Context Specification” is performed. This is because when achieving state transitions while maintaining and inheriting task processing context between different Cells, the goal is not simple data transfer but rather transmitting semantic connections such as the purpose of why analysis was conducted and issues identified from results, while explicitly emphasizing important information. Note that when the subsequent Cell reads the specified context, the Inherited Discourse Exchange that is the purpose of the TIDE protocol is established.

Inherited Discourse Exchange is conducted from six perspectives: Direction, Context, Significance & Priority, Relationships & Dependencies, Abstraction Level, and Execution & Evaluation.

Details of the six perspectives in Inherited Discourse Exchange are shown in Table 2.

1. Direction is set as a perspective that clarifies the ultimate purpose of the task (Why), specific expected outcomes (What), and the main intentions or hypotheses leading to them, forming the foundation for action strategies. This perspective was set based on Rushworth et al.’s (Rushworth et al. 2011) neuroscientific research on reward-guided learning and decision-making in the frontal cortex, and Newell & Simon’s (Newell & Simon 2019) problem-solving theory, realizing action focus through clear goals and intentions, strengthening motivation, and improving planning efficiency.
2. Context is set as a perspective that clarifies background information surrounding the task, including past history related to the task, current situation, key stakeholders, available resources, and constraints. This corresponds to understanding the “stage setting” or

Table 1: TIDE Protocol Six-Step Framework for Cell Transition and Context Inheritance: This protocol defines the standardized procedure for transitioning between Cells while maintaining semantic coherence. Steps 1-3 handle the current Cell’s exit process, including trigger recognition, next Cell identification, and critical context specification. Steps 4-6 manage the state transition declaration and next Cell activation, ensuring inherited discourse exchange is properly established.

Step	Name	Description
Step 1	Transition Trigger	Current Cell determines the necessity for transition by recognizing completion of its responsibilities, specific situational changes, or need for task delegation to another Cell.
Step 2	Next Cell Identification	Identify and select the specific next Cell to transition to based on task nature, complexity, and current context requirements.
Step 3	Inherited Context Specification	Current Cell identifies and organizes critical context information essential for the next Cell’s processing initiation (finalized task objectives, key decisions made, summary of major analysis results, etc.), enabling Inherited Discourse Exchange.
Step 4	State Transition Declaration: End	Current Cell formally declares the end of its role by specifying the ending Cell name, destination Cell name, and critical context information.
Step 5	Next Cell Rule Application	LLM loads the next Cell’s rule file (e.g., *.mdc files in Cursor environment) using appropriate tools (e.g., fetch_rules in Cursor environment) and applies it as the new instruction set, immediately beginning processing according to the new ruleset.
Step 6	State Transition Declaration: Start	The newly activated next Cell declares the start of its role by specifying the starting Cell name and begins processing based on the inherited context.

Table 2: Six Components for Inherited Discourse Exchange: These six perspectives form a comprehensive checklist for context inheritance between Cells. Each perspective is grounded in cognitive science literature: Direction draws on reward-guided learning theory; Context relates to working memory models; Significance & Priority reflects attention and effort theory; Relationships & Dependencies aligns with mental model theory; Abstraction Level connects to metacognition theory; and Execution & Evaluation is based on cognitive load and self-regulated learning theory.

Perspective	Definition
Direction	Task purpose (Why) and expected outcomes (What)
Context	Background information and constraints
Significance & Priority	Priority identification and resource allocation
Relationships & Dependencies	Logical connections and dependencies
Abstraction Level	Appropriate granularity of thinking
Execution & Evaluation	Action execution and progress monitoring

“preconditions” in which the task is placed. This perspective is based on the context control function of the central executive in Baddeley & Hitch’s (Baddeley and Hitch 1974) working memory model, and Miyake & Shah’s (Miyake and Shah 1999) research on the relationship between context processing and working memory in complex cognition, enabling accurate understanding of tasks, setting appropriate judgment criteria, and clarifying implicit assumptions.

3. Significance & Priority is set as a perspective for identifying the most essential and impactful elements for achieving current goals among numerous pieces of information and task elements, and concentrating cognitive resources. This perspective is based on Kahneman’s (Kahneman 1973) capacity limitation theory regarding attention and effort, and Norman & Shallice’s (Norman and Shallice 1986) theory of the supervisory attentional system in attention control and priority setting, realizing efficient allocation of limited cognitive resources, prevention of information overload, and expedited decision-making.
4. Relationships & Dependencies is set as a perspective for grasping the logical “connections” and influence relationships between elements such as individual pieces of information, tasks, and ideas. This corresponds to understanding the “structure” and “interactions” of the element groups that constitute the task. This perspective is based on Johnson-Laird’s (Johnson-Laird 1983) mental model theory regarding structural representation and reasoning processes, and Gentner’s (Gentner 1983) analogy and structural mapping theory regarding relational mapping, enabling overall picture comprehension and structural understanding, improved planning accuracy, root cause identification, and future prediction.

5. Abstraction Level is set as a perspective for appropriately selecting and adjusting the granularity of thinking and information (degree of concreteness and abstraction) according to the situation and purpose. This perspective is based on Flavell’s (Flavell 1979) metacognition theory and the levels of processing model in cognitive psychology (Sternberg and Sternberg 2016), realizing flexible thinking mode selection according to the situation, discovery of essential issues, and effective communication.
6. Execution & Evaluation is set as a perspective for executing specific actions based on formulated plans and strategies, monitoring and evaluating progress and results against goals (set in “Direction”), and modifying actions or plans as needed. This perspective is based on Sweller’s (Sweller 1988) cognitive load theory and Zimmerman’s (Zimmerman 2002) self-regulated learning theory, enabling reliable execution of plans, continuous monitoring of goal achievement, and promotion of learning and improvement cycles from experience.

These six perspectives were discovered inductively during the development process, with theoretical foundations confirmed by consulting cognitive science literature. The six perspectives are not completely independent (orthogonal) but are in a mutually complementary relationship. For example, if “Direction” is not established, judgment of “Significance & Priority” becomes difficult, and understanding “Relationships & Dependencies” deepens the grasp of “Context.” In this study, by treating these perspectives as a comprehensive checklist, we prevent gaps in thinking and achieve precise context inheritance. The exact relationships and weighting of each perspective change dynamically according to the nature of the task and individual cognitive characteristics, and elucidating these is a future research topic.

Design Philosophy: Natural Language EUD, Metacognition, and Iterative Refinement

The CORAL framework is based on the design philosophy of “a mechanism that allows anyone to implement AI adapted to their own thinking processes in natural language and continuously improve it.” This philosophy integrates three elements: (1) end-user development through natural language, (2) promotion of metacognition, and (3) iterative improvement based on feedback.

Functional Classification of Cells: Cells in the CORAL framework are categorized into two types based on their function. First, the **Sensemaker Cell** is a special Cell designed with the intent to promote metacognition (thinking about thinking), designed to pose fundamental questions to users such as “Why?” and “What is the essential problem?” to encourage deep problem understanding rather than superficial problem-solving. Second, **General Cells** (Orchestrator, Planner, Executor, etc.) are implementations of users’ thinking processes that have been structured, verbalized, and implemented as AI reasoning processes. This classification is based on the dual cognitive load strategy in the S-OPE configuration (see Section)—intentional cognitive load increase by Sensemaker and cognitive load reduction by

General Cells—as the design foundation. However, the metacognition promotion effect by Sensemaker is a design intent, and its empirical verification is outside the scope of this study.

Natural Language End-User Development

End-user development (EUD) through natural language serves as a bridge function for converting individual thinking processes into a form executable by AI. In CORAL, users directly describe Cell reasoning processes in natural language without depending on programming languages. This allows domain experts to embed their thinking into the system without technical barriers. While the inherent ambiguity of natural language carries the risk of misinterpretation, CORAL views this ambiguity as an opportunity to clarify intentions through dialogue with AI and convert tacit knowledge to explicit knowledge (see Section).

Metacognitive Enhancement through Description Process

Promotion of metacognition is one of CORAL’s design intents. As Tankelevitch et al. (Tankelevitch et al. 2024) analyzed, in conventional AI use, users are forced to engage in advanced metacognitive activities of prompt formulation, output evaluation, and strategy modification. In CORAL, users structure their thinking patterns as Cell rules. This Cell description act is structurally similar to Flavell’s (Flavell 1979) “metacognition”—“thinking about thinking.” Users gain opportunities to objectively view their own cognitive processes through description. Additionally, by structuring and implementing thinking processes as Cells, it is expected that the load of prompt formulation will be transferred to Cell rules, and output evaluation will change to “consistency confirmation of internalized thinking.”

This design intent is based on Schon’s (Schn 2017) reflective practice theory. CORAL’s Cell design, description, and improvement cycle is designed to support the circulation of “knowledge in action” and “reflection on action.” Users observe AI behavior, recognize gaps with their intentions, and modify Cell descriptions. Unintended AI responses are reframed not as “poor AI performance” but as “ambiguity in thinking process description.” Through this reflective practice loop, users gain opportunities to convert tacit knowledge to explicit knowledge (see Section). However, these metacognition promotion effects are theoretical assumptions based on design intent, and this study has not conducted empirical verification. Longitudinal measurement of effects is a future research topic.

Iterative Refinement Through Feedback Loop

The form realized by the CORAL framework is a feedback loop where users observe AI responses, improve Cells, and observe results again. Cells described in natural language function as a “common thinking blueprint” that is understandable and editable by both humans and AI. Users observe AI responses and improve Cells to adjust AI to a system adapted to their own

cognitive characteristics. At the same time, through dialogue with AI, users themselves gain opportunities to objectively view their own thinking processes and convert tacit knowledge to explicit knowledge.

This improvement cycle can be understood from the perspective of Zimmerman's (Zimmerman 2002) self-regulated learning theory and Argyris & Schon's (Argyris and Schön 1978) organizational learning theory (single-loop learning and double-loop learning). The CORAL framework technically supports users' self-regulated learning processes. Specifically, the act of users designing and implementing their own thinking processes as Cell rules corresponds to the "forethought phase" in self-regulated learning. And the process where the AI agent executes tasks based on implemented Cell rules and users observe those results corresponds to the "performance phase" and "self-monitoring phase."

Users evaluate AI agent responses and artifacts against their own intentions, expectations, and task goals (self-judgment). Based on this evaluation, if there is a gap between expectations and AI agent behavior, users modify and refine Cell rule descriptions, or add or delete new Cells to adjust the AI's reasoning process (self-reaction). This series of user "self-monitoring -> self-judgment -> self-reaction" cycles functions as practical "feedback" for the AI, making AI behavior more adapted to users' thinking styles and task requirements.

The depth of "self-reaction" here determines the quality of improvement. Borrowing from Argyris & Schon's theory, when user reactions remain at fine-tuning Cell rules or clarifying specific instructions, this corresponds to "single-loop learning." This advances improvement within existing frameworks. However, through repeating this single-loop learning, or through unexpected insights gained from dialogue with AI agents, or through the process of articulating one's own thinking processes as Cells (promotion of metacognition), users may arrive at more fundamental questions such as "Is this division of Cell roles really appropriate?" or "Is the current form of information coordination between Cells via the TIDE protocol suitable for essential problem resolution?" At this stage, users may reconsider the design philosophy of existing Cells themselves, add Cells with new roles, or significantly change the coordination structure between Cells and the rules of information inheritance. This corresponds to "double-loop learning," accompanied by transformation of the thinking framework itself.

Thus, the CORAL framework has a structure that can develop from single-loop learning to double-loop learning according to the depth of user reflection, based on the self-regulated learning cycle. Each time users improve Cell rules, the AI agent deploys more refined reasoning according to the updated "thinking blueprint," and the precision and efficiency of task execution improve. Through this continuous feedback loop and occasional fundamental reviews, users enhance their metacognitive ability to objectively view and structure their own thinking processes. As a result, a continuous improvement cycle based on feedback is technically implemented.

The Inherent Difficulty of Thought Process Structuring

The challenge in implementing the CORAL framework is the fundamental difficulty of externalizing and structuring one’s own thinking processes as natural language. This difficulty stems not from superficial issues such as technical implementation complexity or tool usage, but from the essential characteristics of human cognitive processes, particularly attributable to the unconsciousness of thinking processes. Most of human everyday thinking and problem-solving processes are executed automatically and unconsciously (Stanovich and West 2000). For example, processes where experts intuitively grasp problems and find solutions, the basis for experience-based judgments, and criteria for priority decisions often exist as tacit knowledge that even the individuals themselves cannot clearly verbalize. To articulate these thinking processes as Cell rules, metacognitive activity is first required—becoming aware of one’s own unconscious cognitive processes. Properly executing this “thinking about thinking” requires cognitive load and time investment, which can become a psychological barrier for many users.

This barrier paradoxically suggests the value of the CORAL framework. By directly addressing the essential difficulty of metacognition, the CORAL framework finds its *raison d’être* in achieving a deep level of personalization that is difficult to achieve with conventional approaches.

LLM Attention Control Mechanisms

The sections so far have primarily explained the design philosophy and human-centered aspects of the CORAL framework. This section shifts perspective to explain how the CORAL framework functions in terms of the technical aspect of LLM attention mechanisms and achieves stable reasoning processes. This technical understanding provides the foundation for understanding the specific implementation and evaluation of the “S-OPE configuration,” an implementation example of the CORAL framework, in the subsequent Section .

The CORAL framework can be understood as a structure for explicitly and stably executing attention control, utilizing the attention mechanism of Transformer models. In the CORAL framework, complex and advanced custom instruction groups are divided into smaller, manageable functional units called “Cells,” while simultaneously defining and dividing the tasks handled by each Cell. This enables the LLM to focus only on “the task it should handle now” in each Cell, making it easier to maintain attention (Liu et al. 2024).

In the TIDE protocol, rules are established for transmitting particularly important information as “inherited context” when transitioning from one Cell to the next, allowing the LLM to accurately understand “what to focus on next” based on the processing results of the previous Cell, and continue tasks while maintaining attention without losing context. Simultaneously, the TIDE protocol structurally changes the LLM’s reasoning process through dynamic custom instruction loading. Since each Cell defines different thinking patterns (Stage structure, judgment criteria, output format, information processing priorities), the thinking framework itself is switched.

Each Cell’s reasoning process design is a structuring and verbalization of human thinking processes, implemented as AI reasoning processes, processing information, making judgments, and generating outputs according to patterns unique to that Cell. Through this technical implementation, the CORAL framework causes multiple specialized reasoning processes to function in an integrated manner within a single LLM and single session while maintaining semantic coherence.

Based on the explanation of the theoretical foundation and technical characteristics of the CORAL framework presented so far, the next chapter details the S-OPE (Sensemaker-Orchestrator-Planner-Executor) architecture, an implementation example for demonstrating the implementability and effectiveness of this framework. This proof of concept concretely demonstrates how the core concept of the CORAL framework—structuring and implementing individual thinking processes—functions and demonstrates effectiveness in actual research and development situations.

An Implementation: S-OPE

S-OPE Implementation: Overview

In this study, we designed and implemented the S-OPE (Sensemaker-Orchestrator-Planner-Executor) configuration as an implementation example and quantitative evaluation target of the CORAL framework.

The S-OPE configuration is a combination of four types of Cells. The roles of each Cell are shown in Table 3. Sensemaker promotes essential problem inquiry through strategic dialogue; Orchestrator serves as the central coordination hub for task classification and routing; Planner develops systematic execution strategies; and Executor (comprising Task Manager, Writer, Coder, and Runner) handles specialized domain execution.

The conceptual diagram of the S-OPE configuration is shown in Figure 4.

This configuration is a structuring and verbalization of the thinking pattern “essential inquiry -> overall comprehension -> plan formulation -> execution” that the developer cultivated through pharmaceutical research and practical experience in private companies, implemented as AI reasoning processes.

The S-OPE configuration adopts a design philosophy based on cognitive load theory (Sweller 1988). According to Sweller’s cognitive load theory, human working memory has limitations, and the efficiency of learning and problem-solving is determined by appropriate management of cognitive load.

However, in S-OPE, rather than a “cognitive load reduction” approach, a dual strategy is adopted as a design intent: reducing unnecessary cognitive load while intentionally maximizing cognitive load related to essential thinking. Specifically, each Cell—Orchestrator, Planner, and

Table 3: S-OPE Implementation Components: This table presents the four Cell types with their specific implementations and functional roles in the S-OPE configuration for structuring and implementing thinking processes.

Cell Type	Implementation	Functional Role
Sensemaker	00_sensemaker.mdc	Strategic dialogue and essential problem exploration through productive cognitive difficulties, promoting deep reflection and insight generation for complex challenges requiring fundamental reconceptualization
Orchestrator	01_orchestrator.mdc	Central coordination hub implementing separation of concerns principles: instruction interpretation, task classification, initial context gathering, and intelligent routing to appropriate execution paths
Planner	02_planner.mdc	Hierarchical task network planning and systematic execution strategy development based on inherited context from Orchestrator or Sensemaker, employing structured decomposition and dependency analysis
Executor	03_a_task_manager.mdc 03_b_writer.mdc 03_c_coder.mdc 03_d_runner.mdc	Specialized domain execution across four implementations: Task Manager (cross-executor orchestration & progress monitoring), Writer (document creation & editing), Coder (software development), Runner (terminal command execution)

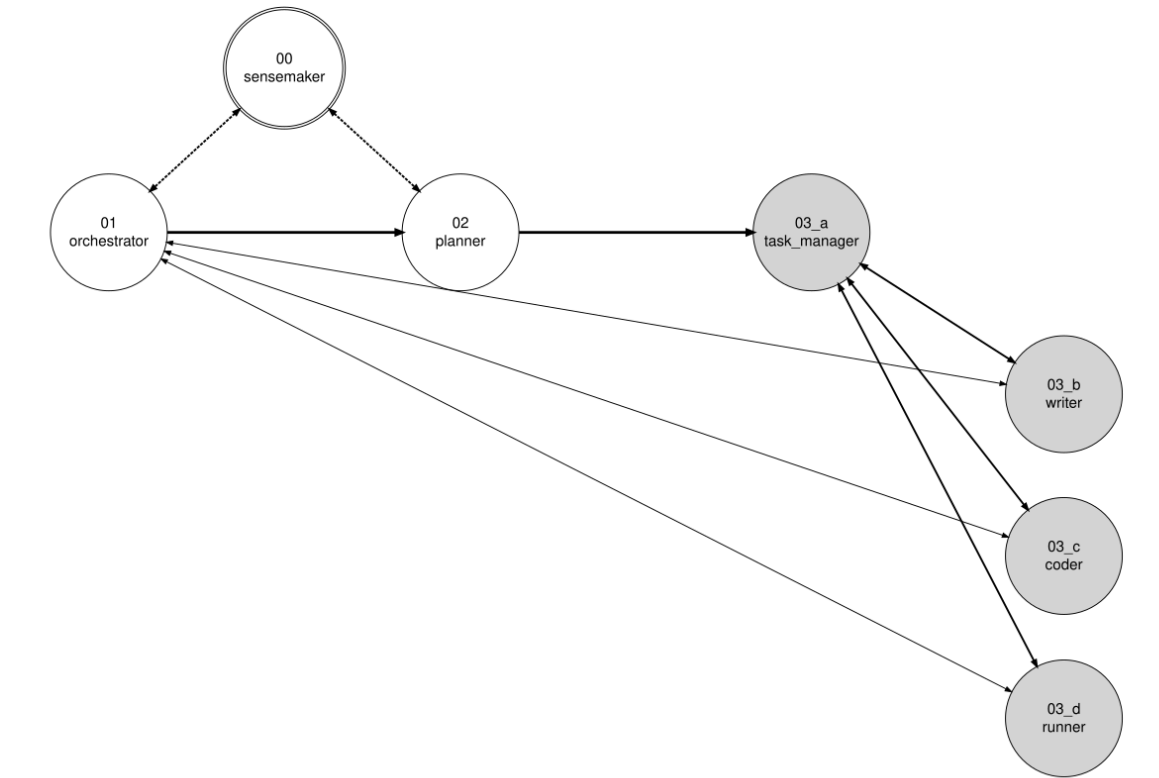


Figure 4: S-OPE Architecture: Four-cell configuration implementing dual cognitive load strategy. Sensemaker (00, double-bordered) provides strategic dialogue promoting metacognition through productive difficulties. The main pathway (Orchestrator->Planner->Task Manager) reduces cognitive load through systematic task decomposition. Task Manager (03_a) orchestrates specialized Executors: Writer (03_b), Coder (03_c), and Runner (03_d), shown in gray. Thick lines indicate primary workflow; dashed lines show optional strategic consultation routes.

Executor—is designed to efficiently reduce cognitive load in task classification, organization, and execution, eliminating the need for users to allocate attentional resources to complex procedures and implementation details. On the other hand, the Sensemaker Cell is designed to intentionally increase cognitive load, requiring fundamental and strategic thinking from users such as “Why?” “What is the essential problem?” and “Are the assumptions appropriate?” with the aim of promoting deep problem understanding rather than superficial problem-solving.

This allocation of cognitive load is a design based on the concept of “desirable difficulties” (Bjork and Bjork 2020) in cognitive science. The strategic dialogue process provided by the Sensemaker Cell is intended as a technical implementation of these desirable difficulties. Note that while the effectiveness of the Sensemaker Cell itself has been statistically confirmed through ablation analysis (see Section), the mechanism of its effect (metacognition promotion, introduction of desirable difficulties, etc.) has not been verified in this study.

S-OPE Implementation: Usecases

Use cases of the S-OPE configuration, extracted from actual usage experience during CORAL development and paper writing, are shown in Table 4. These use cases are practical examples where Cell transition patterns dynamically change according to task complexity and nature, and cognitive load allocation strategies are flexibly applied. For simple tasks, direct execution minimizes procedural cognitive load (Use Case 3); for complex tasks, stepwise planning and execution achieve gradual cognitive load reduction (Use Case 2); for strategic challenges, productive difficulties are intentionally introduced through dialogue with Sensemaker (Use Case 1, 4). The S-OPE configuration is an implementation of the CORAL framework’s “adaptive linkage” concept, tailored to the developer’s thinking processes and cognitively demanding tasks such as framework development and paper writing.

These use cases are actual examples that occurred during the process of writing this paper using the S-OPE configuration.

In concept development, productive difficulties are intentionally introduced through deep dialogue with Sensemaker, promoting essential problem inquiry rather than superficial solutions. The Cell transition pattern including Sensemaker (01->00->01->02->03_a->03_b->03_a->03_d) demonstrates adaptive routing where, when strategic thinking is needed, the Orchestrator recognizes the nature of the task, selects consultation with Sensemaker, gains deep insights, and then transitions to planned execution.

For complex tasks (document writing and technical implementation), systematic cognitive load reduction is achieved through stepwise planning and iterative improvement. This Cell transition pattern (01->02->03_a->03_c->03_a->03_c->03_a->03_b) is commonly used in S-OPE configurations, where Planner and Task Manager collaborate to divide complex tasks into manageable units, and iterative coordination with Coder improves quality. This pattern demonstrates the CORAL framework’s structuring capability to process technical complexity in a form adapted to human cognitive capabilities.

Table 4: S-OPE Use Cases: Cell Transition Patterns Across Different Task Complexities. These patterns demonstrate the CORAL framework’s adaptive linkage capability, where routing dynamically adjusts based on task nature. Simple tasks use direct routes (01->03_b) for efficiency; complex tasks involve planning stages (01->02->03_a->...); strategic challenges engage Sensemaker for essential inquiry (01->00->01->...); and error scenarios trigger hierarchical escalation for robust recovery.

Use Case	Cell Transition Pattern	Key Value
Concept Development	01->00->01->02->03_a->03_b->03_a->03_d	Essential problem exploration through productive difficulties, enabling deep cognitive partnership
Complex Technical Implementation	01->02->03_a->03_c->03_a->03_c->03_a->03_b	Systematic approach to complex tasks with iterative refinement and progress monitoring
Simple Document Modification	01->03_b	Cognitive load reduction through intelligent task classification and direct routing
Error Escalation & Strategic Re-planning	01->02->03_a->03_c(error)->03_a(escalation)->01->00->01	Robust error handling through hierarchical escalation and strategic re-thinking, demonstrating system resilience

For simple document modifications, direct execution (01->03_b) minimizes procedural cognitive load, freeing users from non-essential work. When the Orchestrator recognizes task simplicity, it bypasses the planning stage and directly delegates to Writer, achieving both efficiency and protection of cognitive resources.

Error escalation with strategic re-planning is the most complex pattern demonstrating system robustness, where errors at the execution level are guided to strategic reconsideration through appropriate escalation to higher layers. This demonstrates that the CORAL framework has the capability to adaptively handle unexpected difficulties.

S-OPE Implementation: Development Process

The S-OPE configuration implemented and evaluated in this study was developed for the purpose of proof of concept of the CORAL framework, and the current configuration was established in parallel with framework formation. The S-OPE configuration development process shown below is based on post-hoc analysis of records that naturally accumulated during development, not data collection that was systematically planned. The specific development process of the S-OPE configuration is shown in Table 5.

According to recorded development processes, implementation began in January 2025, and quantitative expansion from 1 rule (3.4K characters) to 7 rules (38K characters) was observed over approximately 5 months. Major technical changes at each stage recorded include: introduction of strategic dialogue functionality (April 16, 2025), addition of orchestration functionality (April

Table 5: S-OPE Development Process: Scale Progression (1->7 rules) and Architectural Evolution Over Five Months. The development shows iterative refinement from a single coding rule to a complete seven-rule framework. Key milestones include: strategic dialogue introduction (April 2025), orchestration addition (April 2025), integration experiment failure leading to Cell structure completion with TIDE protocol emergence (May 2025), and complete TIDE protocol with six-perspective inherited context definition (June 2025).

Development Phase	Configuration Scale	Key Technical Innovation	Architectural Evolution
Initial (Jan 2025~)	1 rule, 3.4K chars	Basic coding	Single-purpose tool
2025-04-16 (Dialogue)	2 rules, 6.5K chars	Strategic dialogue Multi-instruction	Dual-mode coordination
2025-04-27 (Orchestration)	4 rules, 12K chars	Dynamic switching Cell-like architecture	Multi-agent orchestration
2025-04-28 (Integration)	3 rules, 15K chars	Integration attempt (stability reduced)	Failed integration -> learning
2025-05-09 (Cell Structure)	7 rules, 36K chars	TIDE protocol emergence Cell structure complete (inheritance undefined)	Modular recovery Near-current form
2025-06-05 (TIDE Complete)	7 rules, 38K chars	6-perspective context TIDE complete	Stable final system

27, 2025), integration experiment and its failure (April 28, 2025), Cell structure completion with TIDE protocol emergence (May 9, 2025), and complete TIDE framework implementation (June 5, 2025).

In particular, during the period from April 28, 2025 to May 9, 2025, after the failure of the integration experiment, the TIDE protocol first appeared and the Cell structure was completed. During this period, the description volume increased approximately 2.5 times from 15K characters to 36K characters. However, the six perspectives of inherited context were not yet defined at this point.

The final definition of the six perspectives was completed in the June 5, 2025 version, establishing the complete TIDE protocol and the S-OPE configuration that became the target of quantitative evaluation.

The following Section describes the statistical experimental design for quantitatively evaluating what performance the S-OPE configuration actually demonstrates on complex tasks.

Evaluation Framework

Research Design and Comparative Methodology

This study adopts a first-person research approach where the developer serves as the evaluator. In the HCI field, when designing and evaluating novel interactive systems, introspective insights gained from the designer’s deep engagement with the system are indispensable (Schn 2017). Particularly given the nature of the CORAL framework—structuring and implementing individual thinking processes—a proof of concept targeting the developer’s own cognitive processes is the most direct method for demonstrating framework feasibility. This methodological choice brings constraints on external validity (generalizability) (see Section), but is a valid approach for proof of concept in unexplored areas, paving the way for subsequent research verification with diverse users.

In this study, we conducted controlled quantitative evaluation to verify the effectiveness of the CORAL framework within the framework of statistical hypothesis testing.

All tasks in this evaluation were conducted on Cursor (ver. 1.1.3 to 1.2.1), using a single LLM (claude-4-sonnet: thinking model, released 2025-05-22). Features that could affect AI agent behavior in the Cursor IDE were restricted. Specifically, Rules for AI (special custom instructions effective for all projects) contained only instructions regarding response language (Always respond in Japanese), and Memories (a function to save and rule-ify useful context from chat) and To-Do List (a function to generate task lists for AI agents to track tasks) were prohibited.

A complex high-difficulty task was set as the challenge, and reports (final artifacts) were scored through rubric evaluation by a dedicated scoring AI agent, comparing score distributions. The overall design of quantitative evaluation is shown in Figure 5.

Evaluation Tasks and Complexity Design

In this study, the “report” as the final artifact from high-difficulty tasks was scored using a pre-established rubric.

Task: Business Challenge

Designed as a problem discovery and plan modification task in a new product launch plan, evaluating the following capabilities: (a) ability to formulate structured plans from ambiguous initial instructions, (b) ability to respond to severe constraint conditions added midway (budget cuts) and make realistic plan modifications, and (c) ability to autonomously detect contradictions in presented information (conflict between mandatory functional requirements and legal regulations), raise issues, and propose logical examination and solutions.

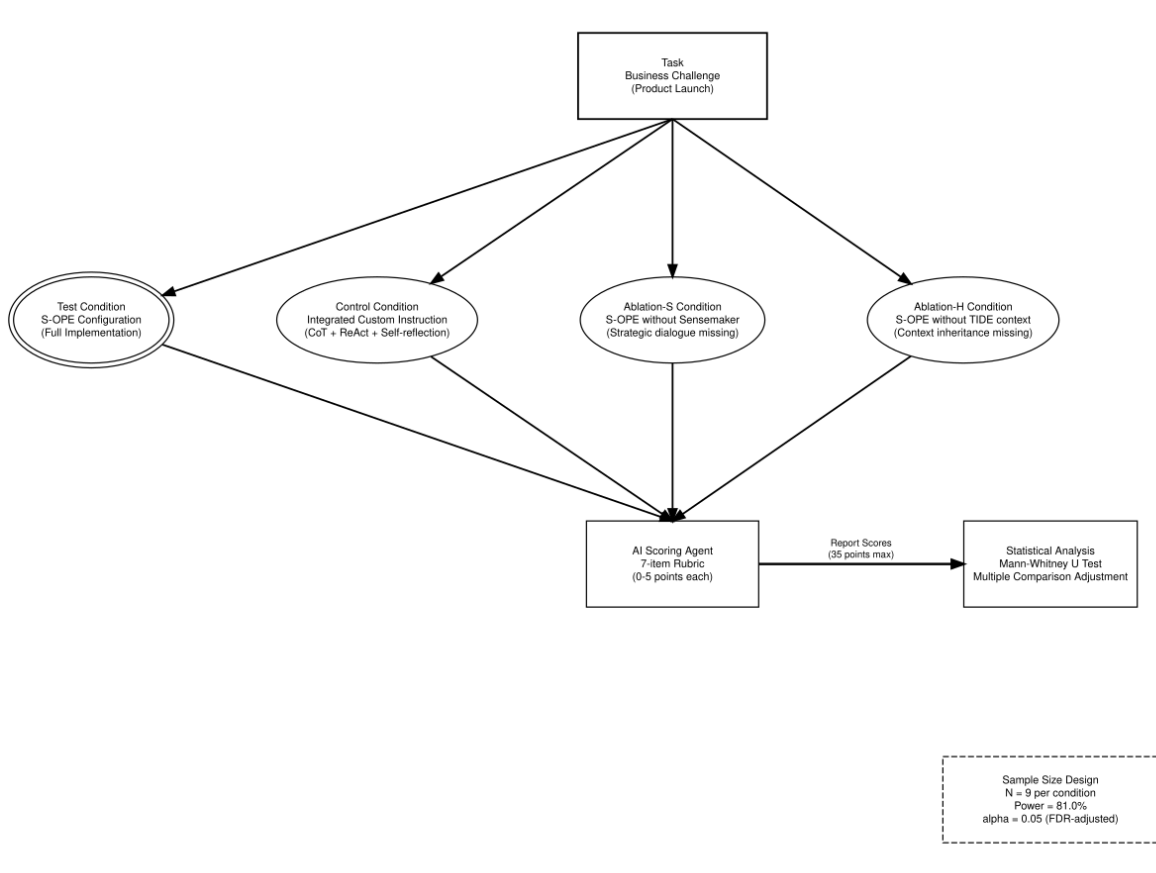


Figure 5: Experimental Design: Four-condition comparative study evaluating the CORAL framework effectiveness. Test condition uses the complete S-OPE configuration; Control condition uses integrated custom instructions combining CoT (Chain-of-Thought), ReAct (Reasoning and Acting), and Self-reflection; Ablation-S removes the Sensemaker Cell to evaluate strategic dialogue contribution; Ablation-H removes TIDE context inheritance definitions to evaluate context inheritance contribution. Each condition includes N=9 trials, evaluated by a scoring AI agent using a 7-item rubric (35 points maximum). Statistical power was set at 81.0% with $\alpha=0.05$ and FDR (false discovery rate) adjustment.

In the scenario context, extensive deep information obtainable through questioning is set, designed so that appropriate information extraction enables achievement of 5-point evaluation. Following the information disclosure protocol, information is progressively disclosed according to AI response level and dialogue depth.

Information Disclosure Protocol

The above tasks are intentionally set as high-difficulty tasks that directly test advanced cognitive processes such as “progressive complex information inheritance,” “long-term contextual consistency maintenance,” and “problem redefinition and refinement through dialogue.” This aims to evaluate the CORAL framework’s characteristics from multiple angles and faithfully simulate real-world complex problem-solving processes. To improve evaluation reproducibility and fair evaluation of AI capabilities, the following hierarchical information disclosure protocol was adopted for tasks in this study:

Level 1 (Basic Information): Basic facts, background information, term definitions, etc. necessary for task execution. Disclosed when AI makes direct, specific questions about particular facts or basic situations.

Level 2 (Analytical Information / Background Issues): Underlying more complex issues, potential conflicts of interest between stakeholders, detailed situations that may influence decision-making, etc. Disclosed when AI goes beyond information exploration to connect multiple pieces of information for situational analysis, presents questions or hypotheses based on deeper insights, or asks about hidden assumptions or causal relationships beyond presented information.

Level 3 (Core Information / Deep Psychology / Strategic Information): Important undisclosed information related to task essence, deep psychology and true intentions of characters, confidential information affecting strategic judgment, important constraint conditions intentionally concealed, etc. Disclosed when AI attempts to identify root causes of problems, integrates complex multiple pieces of information to suggest essential problem-solving directions, asks questions that approach the core of ethical dilemmas, or infers the true motives and intentions behind characters’ actions and statements—demonstrating extremely deep insight, analysis, and questioning.

Evaluation Rubrics and Scoring Protocol

Seven-Item Evaluation Rubric

Reports were evaluated on the following 7 items, scored on a 35-point maximum scale with 0-5 points for each item:

1. **Essential Problem Identification and Structuring Capability:** Ability to capture essential problem structure from multiple angles in complex information and situations, clearly define problem cores, and effectively redefine problems to be solved when necessary.
2. **Adaptive Analysis Capability:** Ability to flexibly adjust and deepen analysis and proposals in response to changing constraints (budget, new information, etc.).
3. **Multi-perspective Consideration Capability:** Ability to gain insights from diverse perspectives (technical, ethical, social impact, etc.) and consider potential risks and biases.
4. **Integrated Impact Evaluation Capability:** Ability to evaluate multi-faceted impacts of proposals and choices (ethical, planning, stakeholder impacts, etc.), consider trade-offs, and support rational decision-making.
5. **Practical Solution Proposal Capability:** Ability to deeply understand task background and context (stakeholders, constraints, objectives, etc.) and formulate and propose specific, executable solutions.
6. **Value-Aligned Output Capability:** Ability to produce final proposals and artifacts that are aligned with basic project/organization purpose values (mission, ethics, etc.) and are logical and high-quality.
7. **Logic and Information Reliability:** Generated reports are described plainly and accurately based on clear logical structure, with reliability and validity of referenced information sources guaranteed.

Each item was scored on a 0-5 point scale according to detailed meta-criteria (0: Insufficient, 1: Needs Improvement, 2: Partial, 3: Good/As Expected, 4: Excellent, 5: Outstanding). The complete rubric definitions including scoring criteria for each level are available in the Zenodo repository (k-shibuki 2025).

Scoring AI Agent Operating Protocol

To minimize evaluation bias, scoring was conducted by a scoring AI agent designed in advance for each task, following this protocol:

- **Evaluation Session Independence:** To eliminate bias from evaluation history, evaluation starts in a new chat session.
- **Information Restriction:** Only the “Report” to be evaluated and related task definitions are provided as information.
- **Condition Blinding:** Condition identifiers use symbols (t, c, as, ah) to prevent the scoring AI agent from inferring condition meaning from names. Combined with the information restriction above, the scoring AI agent evaluates reports without knowledge of which experimental condition produced them, achieving single-blind evaluation with respect to the AI evaluator.
- **Three Evaluations with Average Value Adoption:** Exactly 3 evaluations are conducted for each evaluation target, and the average value is used as the final artifact score. Average values are processed to the second decimal place according to IEEE 754, and the first decimal place value is adopted.

- Transparency Assurance: Chat history of all evaluation processes (evaluation chat logs) is recorded and published unedited.

Comparison/Control Conditions

The following four conditions were set for systematic comparative evaluation. In designing Ablation conditions, we targeted two core functions of the CORAL framework: the strategic dialogue function by Sensemaker Cell and the context inheritance function by TIDE protocol. The reason for selecting these two rather than other Cells such as Orchestrator or Planner is that Sensemaker is a representative example of “functions impossible to realize with a single prompt” (see Section), and the TIDE protocol is an essential mechanism for maintaining semantic coherence after Cell division. That is, these two are components that directly support the *raison d’être* of the CORAL framework, and we judged that identifying their contributions was most important for verifying framework effectiveness.

1. Test Condition: As S-OPE Configuration, uses the complete implementation of the CORAL framework composed of multiple advanced custom instruction groups (Sensemaker, Orchestrator, Planner, Executor).
2. Control Condition: As Integrated Custom Instruction, uses a single advanced custom instruction integrating CoT (Chain of Thought), ReAct (Reasoning and Acting), and Self-reflection elements. This condition functions as a representative of existing advanced prompt engineering techniques, providing a strong baseline. Specifically, it is a custom instruction of approximately 2,000 characters including a 5-stage structured thinking process, 7 self-evaluation items, explicit requirements for critical thinking (bias detection, logical leap elimination), and a 3-part fixed output format—a design with 4-5 times the technical density compared to typical official guidelines (approximately 200-500 characters). Note that the Control condition does not include strategic dialogue functionality like that provided by the Sensemaker Cell. This is not an arbitrary omission but due to technical constraints. If the Sensemaker Cell (approximately 4,500 characters) were integrated into the Control condition (approximately 2,000 characters), it would become a single custom instruction of approximately 6,500 characters or more, causing attention control breakdown (see Section). This technical constraint suggests the *raison d’être* of the CORAL framework—enabling functions impossible with a single prompt through Cell division.
3. Ablation-S Condition: As S-OPE Configuration (Sensemaker Cell Missing), uses the S-OPE configuration with the Sensemaker Cell removed. This identifies the contribution of strategic dialogue for essential inquiry.
4. Ablation-H Condition: As S-OPE Configuration (Handover Context Definition Missing), uses the S-OPE configuration with all custom instruction descriptions related to `{{handover_context}}` definition and use between Planner Cell and Executor Cell removed. This identifies the contribution of context inheritance functionality by the TIDE protocol.

Counterbalancing Using Latin Square Design

To minimize order effects such as learning and fatigue effects arising from experimenter prompt input during task execution, the implementation order of the four conditions (Test, Control, Ablation-S, Ablation-H) was counterbalanced using a Latin square design. Specifically, each condition was assigned to different implementation positions (1st through 4th) as equally as possible across the entire trial set of $N=36$ ($9 \text{ trials} \times 4 \text{ conditions}$).

Statistical Analysis and Power Analysis

Primary Endpoint

This study used report score distribution as the primary evaluation indicator, statistically testing whether there are significant differences in the distribution of 7-item evaluation scores (0-5 points per item, 35 points total) scored by the scoring AI agent based on the rubric.

Secondary Endpoints

1. Median of Excellent Evaluations (4-point) Achieved in Final Artifact Scores: Compared the median of the number of 4-point scores achieved among the 7 items of final artifact scores between conditions.
2. Median of Outstanding Evaluations (5-point) Achieved in Final Artifact Scores: Compared the median of the number of 5-point scores achieved among the 7 items of final artifact scores between conditions.

Statistical Methods and Sample Size Design

For statistical comparison, Mann-Whitney U test (Wilcoxon rank-sum test) was adopted considering small samples and ordinal scales. This non-parametric method does not assume normal distribution, is robust to outliers, and functions appropriately with small samples. It tests differences in medians and provides practical and easy-to-interpret comparisons.

Power Analysis for Optimal Sample Size Determination

For controlled statistical comparison, simulation-based power analysis was implemented in advance. Statistical power (Type II error probability) was pre-calculated to determine appropriate sample size. The goal was to detect a median improvement of 3 points (assumed median of 28 points) compared to the Control condition’s “final artifact score” (assumed median of 25 points) as the minimum practically meaningful effect. This 3-point improvement corresponds to the S-OPE configuration achieving an average of 4 points (Excellent) in a task where the

Control condition’s final artifact score average is 3.57 points (4 items at 4 points, 3 items at 3 points), representing a non-negligible difference in actual problem-solving.

The following assumptions were set in power analysis:

- Primary Analysis: Equal variance ($\sigma=2.0$) was assumed for both groups—Control group (mean 25 points, standard deviation 2.0 points) and S-OPE group (mean 28 points, standard deviation 2.0 points)—using a model that rounds values generated from normal distribution to integers and clips within range.
- Sensitivity Analysis: Using truncated normal distribution to impose more realistic constraints. Control group (mean 25 points, standard deviation 2.0 points, truncation range [18, 30]), S-OPE group (mean 28 points, standard deviation 2.0 points, truncation range [21, 33]).

Through 10,000 simulations, it was confirmed that $N = 9$ per group achieves 81.0% power (80.5% in sensitivity analysis), exceeding the target power of 80%. Comprehensively considering this result and research feasibility, the final sample size was determined to be $N = 9$ per group. The complete power analysis implementation including R scripts is available in the Zenodo repository (k-shibuki 2025).

Multiple Comparison Adjustment

For the 9 statistical hypothesis tests included in primary analysis (1 Primary Endpoint x 3 conditions) and secondary analysis (2 Secondary Endpoints x 3 conditions), False Discovery Rate (FDR) was controlled using the Benjamini-Hochberg method. 95% confidence intervals were reported for all statistical hypothesis tests, with the second decimal place processed according to banker’s rounding (round half to even) and reported.

Transparency and Reproducibility Measures

To ensure research transparency and reproducibility, the following comprehensive measures were implemented. Please refer to Section for details.

- Pre-publication of Protocol: This evaluation protocol was registered in a public repository and a DOI (Digital Object Identifier) was obtained before data acquisition began, and published.
- Complete Data Publication: Evaluation task definition files, custom instructions (including rubric-equipped scoring AI agent), all chat logs, and evaluation chat logs were published unedited. All research data, statistical analysis code, and the original Japanese implementation of custom instructions are permanently accessible through the Zenodo repository (k-shibuki 2025).

Table 6: Descriptive Statistics of Final Artifact Scores by Condition: Test (S-OPE) demonstrates the highest median score (30.7) compared to Control (25.0), Ablation-S (27.3), and Ablation-H (26.3). SD = Standard Deviation.

Condition	N	Median	Mean	SD	Min	Max
Test (S-OPE)	9	30.7	30.7	1.20	29.0	32.7
Control	9	25.0	24.8	1.57	22.0	27.0
Ablation-S	9	27.3	27.4	1.23	25.0	29.0
Ablation-H	9	26.3	26.3	1.30	25.0	28.3

- Statistical Analysis Reproducibility Assurance: For statistical analysis using R (programming language), versions of R and major libraries used are clearly stated in the paper, and power analysis implementation, all experimental data, and all custom instructions used in S-OPE implementation are published. The protocol was pre-published prior to research implementation, and the entire evaluation process was published unedited for transparency assurance.

Note that all custom instruction implementations including the S-OPE configuration were conducted in Japanese. This language choice is a necessary decision given the core concept of the CORAL framework: “structuring and implementing individual thinking processes.” This is because the developer’s thinking processes have been formed and developed in Japanese, their native language, and using Japanese when structuring, verbalizing, and implementing those thinking processes in AI is thought to maximally preserve the essential structure and semantic nuances of the thinking. For international readers, English translations of Test condition, Control condition, and scoring AI agent are provided in the public repository. Please refer to Section for this point.

Results

Descriptive Statistics

Table 6 shows the descriptive statistics of final artifact scores for each condition. The Test condition (S-OPE configuration) showed a median of 30.7 points, achieving the highest score compared to Control condition (25), Ablation-S condition (27.3), and Ablation-H condition (26.3).

The intra-rater consistency of the AI scoring agent was high, with a mean SD of 1.74 points (range: 0–4.36) across the 36 reports’ three evaluations per report.

Figure 6 shows the distribution of final artifact scores for each condition as a box plot. The Test condition shows a clearly higher score distribution compared to all other conditions, with visually confirming minimal overlap in distributions.

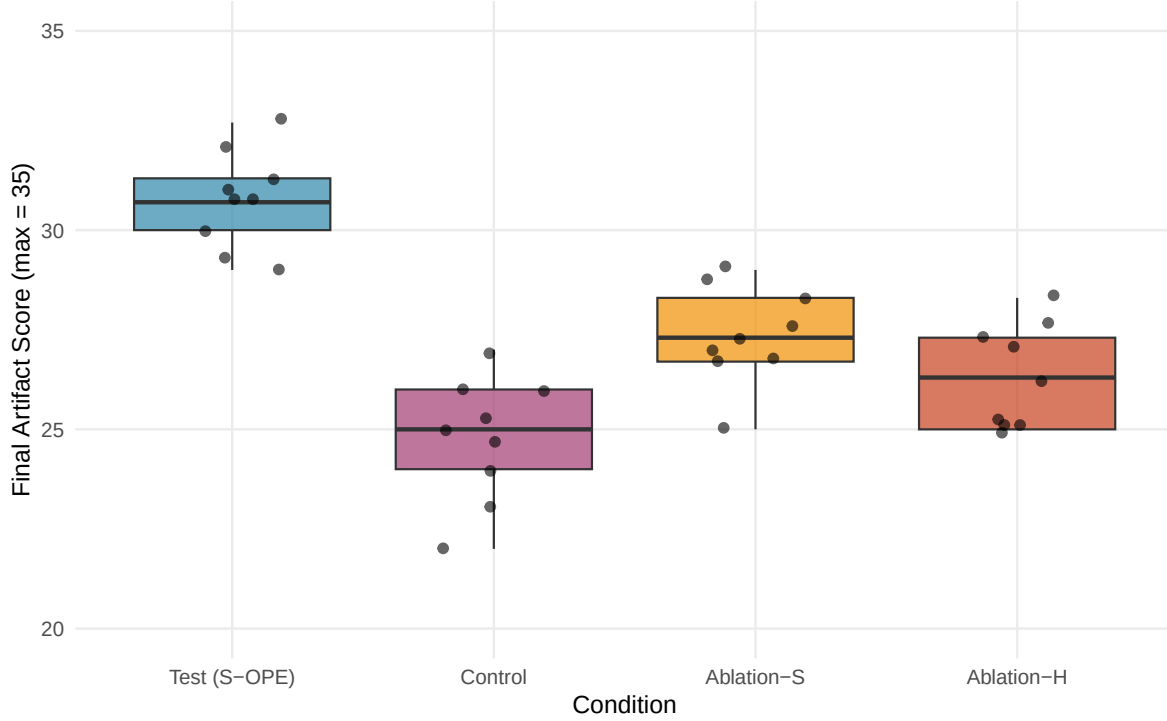


Figure 6: Distribution of Final Artifact Scores by Condition: Box plot showing score distributions across four experimental conditions. The box represents the interquartile range (IQR, Q1-Q3), the horizontal line within the box indicates the median, and whiskers extend to the minimum and maximum values within 1.5xIQR. Test (S-OPE) demonstrates consistently higher scores with minimal overlap with other conditions. Statistical comparisons were performed using Mann-Whitney U tests (two-sided, $\alpha = 0.05$) with Benjamini-Hochberg FDR correction for multiple comparisons.

Order Effects Verification

Figure 7 shows the relationship between execution order and final artifact scores for each condition. This visualization confirms the absence of systematic order effects (learning or fatigue effects) that the Latin square counterbalancing was designed to control.

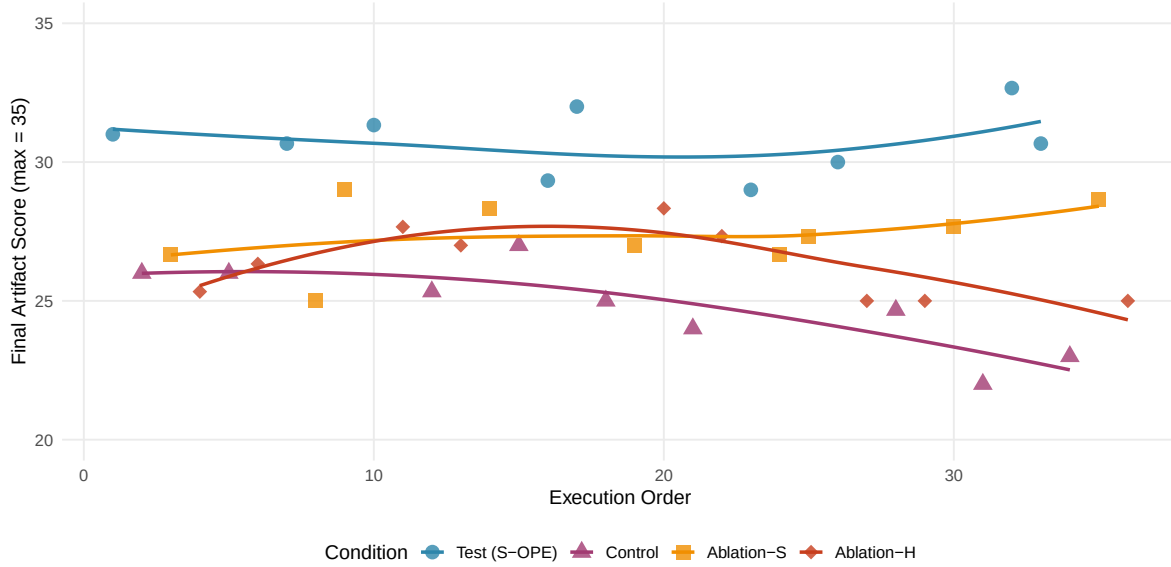


Figure 7: Verification of Order Effects: Scatter plot showing final artifact scores against execution order for each condition. Points are colored by condition and connected with LOESS (Locally Estimated Scatterplot Smoothing) lines to reveal any systematic trends. The absence of consistent upward (learning effect) or downward (fatigue effect) trends across conditions indicates that the Latin square counterbalancing successfully controlled for order effects.

The scatter plot with LOESS smoothing lines reveals no systematic order effects across conditions. If learning effects (score improvement over time) or fatigue effects (score decline over time) were present, all conditions would show consistent upward or downward trends, respectively. Instead, the observed patterns differ by condition: the Test condition maintains stable scores around 30 points throughout the study period, Ablation-S and Ablation-H show relatively flat trends, and the Control condition shows a slight downward slope. Importantly, this variation in trends across conditions—rather than a uniform trend—indicates that the Latin square counterbalancing successfully controlled for order effects. The slight downward trend observed in the Control condition is attributable to the inherent variability of that condition rather than a systematic fatigue effect, as other conditions do not exhibit similar patterns. Therefore, the condition differences observed in the primary analysis can be attributed to the experimental conditions themselves rather than to order effects.

Outstanding Evaluation Acquisition Patterns

Figure 8 shows the pattern of achieving 5-point (Outstanding) evaluations across the seven evaluation dimensions by condition. This visualization reveals which dimensions are more likely to achieve outstanding evaluations and how the patterns differ between conditions.

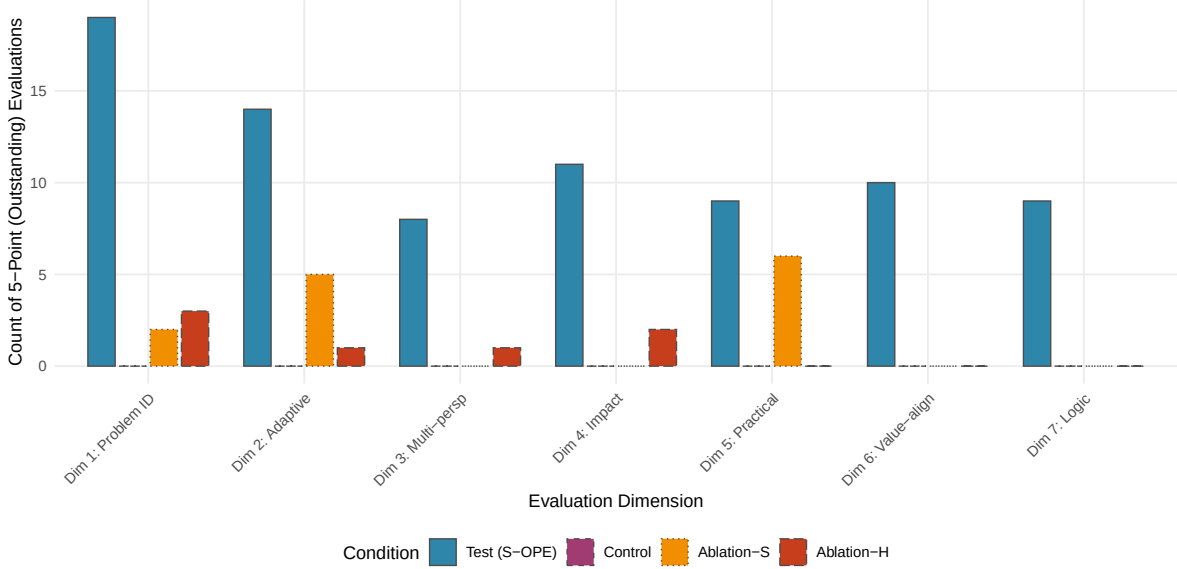


Figure 8: Outstanding (5-Point) Evaluation Acquisition Patterns by Dimension and Condition: This figure shows the frequency of achieving 5-point evaluations across the seven evaluation dimensions (Dim = Dimension) for each condition. The Test condition (S-OPE) achieves outstanding evaluations consistently across all dimensions. The Control condition shows almost no 5-point evaluations. The Ablation conditions achieve some 5-point evaluations in specific dimensions: Ablation-S primarily in Dimensions 2 (Adaptive) and 5 (Practical), and Ablation-H primarily in Dimensions 1 (Problem ID) and 4 (Impact).

The visualization reveals that the Test condition achieved 5-point evaluations consistently across all seven dimensions, with particularly high frequencies in Dimension 1 (Essential Problem Identification), Dimension 2 (Adaptive Analysis), and Dimension 6 (Value-Aligned Output). Notably, while the Control condition achieved almost no 5-point evaluations, the Ablation conditions achieved some: Ablation-S (TIDE present, Sensemaker absent) showed 5-point evaluations primarily in Dimensions 2 and 5, while Ablation-H (Sensemaker present, TIDE absent) showed them primarily in Dimensions 1 and 4. This pattern suggests that each component contributes independently to achieving outstanding evaluations in specific dimensions, but only the full S-OPE configuration achieves consistent excellence across all dimensions.

Table 7: Primary Analysis Results: Mann-Whitney U Test (Two-Sided) Comparing Test (S-OPE) vs. Each Comparison Condition. All comparisons show statistically significant differences with very large effect sizes ($r > 0.8$ per Cohen’s criteria). The 95% confidence intervals for location shift and common language (CL) effect sizes (probability of superiority) are provided. W represents the Mann-Whitney U statistic.

Comparison	W	Raw p	95% CI	Effect Size (r)	CL Effect Size
Test vs Control	81.0	0.0004	[4.3, 7.4]	0.83	1.00
Test vs Ablation-S	80.5	0.0005	[2.0, 4.6]	0.82	0.99
Test vs Ablation-H	81.0	0.0004	[3.0, 5.7]	0.83	1.00

Table 8: Secondary Analysis: Comparison of 4-Point (Excellent) and 5-Point (Outstanding) Evaluation Counts. This table presents the median counts of high evaluations per condition (left columns) and Mann-Whitney U test results comparing Test vs each condition (right columns). The Test condition shows consistently high 5-point evaluation counts (median = 3.0), while other conditions have median = 0.0 despite achieving some 5-point evaluations in specific dimensions. The lower 4-point count for Test reflects elevation to 5-point rather than deficiency.

Condition	Median 4-pt	Median 5-pt	vs Test (4-pt)	vs Test (5-pt)
Test (S-OPE)	3.3	3.0	-	-
Control	4.0	0.0	p=0.894	p<.001*
Ablation-S	5.7	0.0	p=0.011	p<.001*
Ablation-H	4.7	0.0	p=0.082	p<.001*

Primary Analysis: Mann-Whitney U Test

Mann-Whitney U tests (two-sided) were conducted on final artifact scores between the Test condition and each comparison condition. Results are shown in Table 7.

In all comparisons, the Test condition showed statistically significantly higher scores (all $p < 0.001$). Effect sizes (Wilcoxon r) all exceeded 0.8, classified as “very large effects” by Cohen’s criteria.

Secondary Analysis: Excellent and Outstanding Evaluations

As Secondary Endpoints, the number of 4-point (Excellent) and 5-point (Outstanding) evaluations achieved in the 7-item evaluation were compared between conditions. Results are shown in Table 8.

In achieving 5-point (Outstanding) evaluations, the Test condition showed a median of 3, demonstrating a notable difference compared to Control (0), Ablation-S (0), and Ablation-H

Table 9: Multiple Comparison Adjustment Using Benjamini-Hochberg Method: This table presents false discovery rate (FDR) controlled p-values and 95% confidence intervals for all 9 statistical tests (3 primary endpoint comparisons + 6 secondary endpoint comparisons). All primary endpoint comparisons and all 5-point evaluation comparisons remain statistically significant after FDR adjustment (adjusted $p < 0.05$, marked with *). The 4-point evaluation comparisons were not significant, consistent with the Test condition achieving 5-point rather than 4-point evaluations.

Test	Raw p	Adjusted p	95% CI	Significant
Primary: Test vs Control	0.0004	0.0009	[4.3, 7.4]	*
Primary: Test vs Ablation-S	0.0005	0.0009	[2.0, 4.6]	*
Primary: Test vs Ablation-H	0.0004	0.0009	[3.0, 5.7]	*
4-point: Test vs Control	0.8938	0.8938	[-1.7, 1.3]	
4-point: Test vs Ablation-S	0.0113	0.0145	[-2.7, -0.3]	*
4-point: Test vs Ablation-H	0.0824	0.0926	[-2.3, 0.3]	
5-point: Test vs Control	0.0002	0.0009	[2.7, 3.7]	*
5-point: Test vs Ablation-S	0.0009	0.0014	[1.3, 3.3]	*
5-point: Test vs Ablation-H	0.0003	0.0009	[1.7, 3.7]	*

(0). This result suggests that the S-OPE configuration is qualitatively different in its ability to achieve outstanding evaluations, not just achieving higher average scores.

Multiple Comparison Adjustment

FDR adjustment using the Benjamini-Hochberg method was performed on a total of 9 tests from Primary Analysis (3 tests) and Secondary Analysis (6 tests). Results are shown in Table 9.

Even after FDR adjustment, statistical significance was maintained for all Primary Endpoint comparisons (Test vs Control, Test vs Ablation-S, Test vs Ablation-H) and all 5-point evaluation comparisons (adjusted $p < 0.001$). No significant difference was observed for 4-point evaluations, but this is attributable to the Test condition achieving 5-point rather than 4-point evaluations more frequently.

Table 10: Exploratory Analysis: Score Distributions by Evaluation Dimension and Condition. This table presents the median and interquartile range (IQR) for each of the seven evaluation dimensions (0-5 scale) across all four conditions. The Test condition shows consistently higher median scores across all dimensions. Note: This is exploratory analysis for hypothesis generation; no statistical tests were performed on individual dimensions.

Dimension	Test	Control	Ablation-S	Ablation-H
1. Problem Identification	5.0 (4.0–5.0)	3.0 (3.0–4.0)	4.0 (4.0–4.0)	4.0 (4.0–4.0)
2. Adaptive Analysis	5.0 (4.0–5.0)	4.0 (4.0–4.0)	4.0 (4.0–4.0)	4.0 (4.0–4.0)
3. Multi-perspective	4.0 (4.0–5.0)	3.0 (3.0–3.0)	4.0 (3.0–4.0)	3.0 (3.0–4.0)
4. Impact Evaluation	4.0 (4.0–5.0)	4.0 (3.0–4.0)	4.0 (4.0–4.0)	4.0 (3.5–4.0)
5. Practical Solution	4.0 (4.0–5.0)	4.0 (4.0–4.0)	4.0 (4.0–4.0)	4.0 (4.0–4.0)
6. Value Alignment	4.0 (4.0–5.0)	3.0 (3.0–4.0)	4.0 (3.0–4.0)	4.0 (3.0–4.0)
7. Logic & Reliability	4.0 (4.0–5.0)	4.0 (3.0–4.0)	4.0 (4.0–4.0)	4.0 (3.0–4.0)

Exploratory Analysis: Item-Specific Score Distributions

As exploratory analysis, Table 10 shows the score distributions (median and IQR) for each of the seven evaluation dimensions by condition. This descriptive comparison reveals which dimensions show the largest differences between conditions.

Figure 9 visualizes the score distributions as a heatmap, showing the frequency of each score (0-5) for each dimension and condition.

The exploratory analysis reveals that the Test condition achieved higher median scores across all seven dimensions compared to other conditions. The heatmap shows that the Test condition has a concentration of scores at 4-5 points, while other conditions show more variability with scores distributed across 3-4 points. Notably, the IQR patterns reveal different stability characteristics: the Test condition shows consistent IQRs of 4.0–5.0 across dimensions, indicating stable high performance, while Ablation-S showed zero variance (IQR = 4.0–4.0) in most dimensions (5 of 7) and Ablation-H in some (3 of 7), suggesting a “ceiling effect” at 4 points where they cannot break through to outstanding evaluations. This pattern is consistent with the primary analysis results and suggests that the S-OPE configuration’s effectiveness is not limited to specific evaluation aspects but extends broadly across multiple cognitive dimensions.

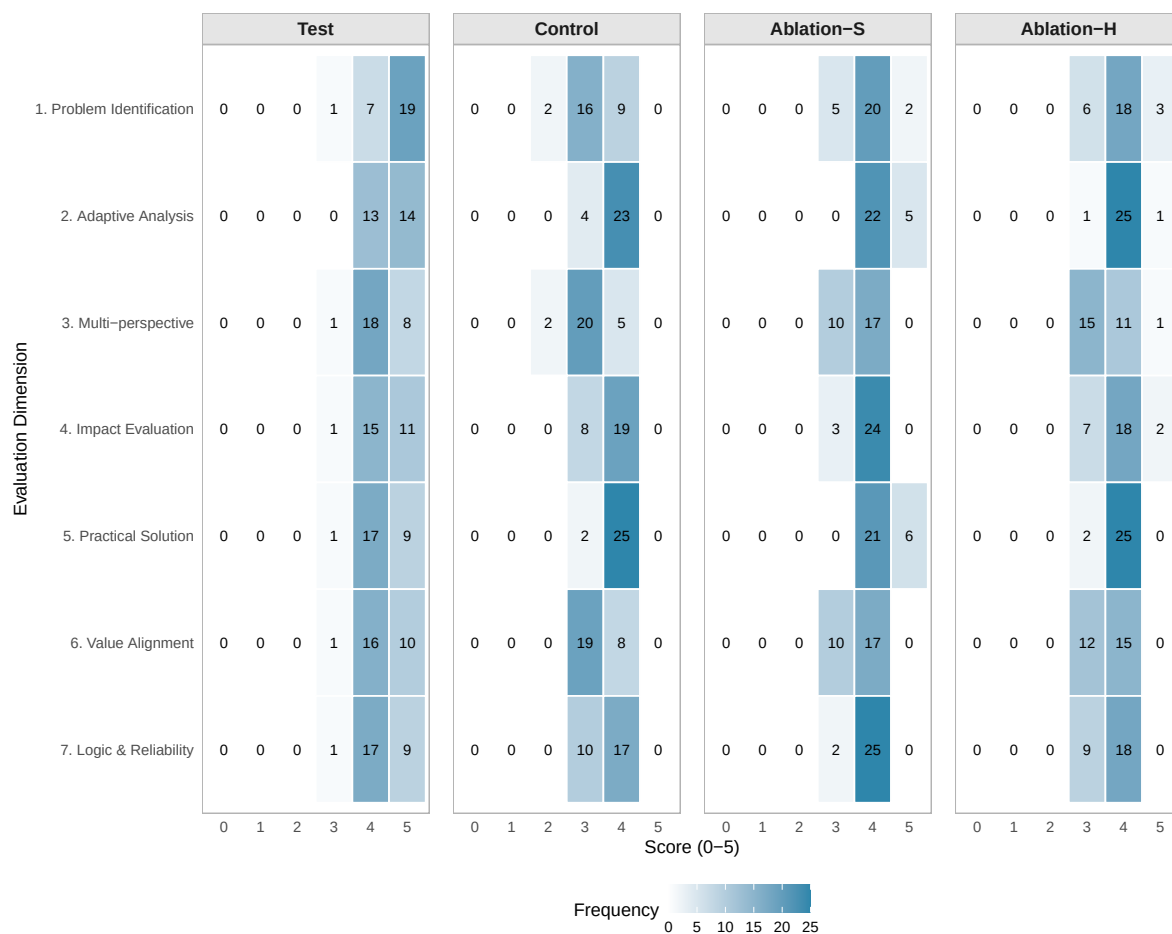


Figure 9: Score Distribution Heatmap by Dimension and Condition: Each cell shows the frequency of scores (0-5 points) for each evaluation dimension. Darker colors indicate higher frequencies. The Test condition shows a concentration of scores at 4-5 points across dimensions, while other conditions show more scores at 3-4 points.

Summary of Results

The following key findings were obtained from the quantitative evaluation in this study:

1. **Primary Endpoint:** The S-OPE configuration (Test condition) showed statistically significantly higher scores compared to all comparison conditions in final artifact scores (FDR-adjusted $p < 0.001$, effect size $r > 0.8$).
2. **Comparison with Control Condition:** The S-OPE configuration showed a median difference of 5.7 points (30.7 vs 25) compared to the Control condition integrating existing advanced prompt engineering techniques (CoT + ReAct + Self-reflection).
3. **Ablation Analysis:** Removal of both Sensemaker Cell (strategic dialogue function) and TIDE protocol (context inheritance function) resulted in significant score decreases. This indicates that both functions of the CORAL framework independently contribute to effectiveness.
4. **Qualitative Superiority:** The Test condition achieved consistently high 5-point (Outstanding) evaluations across all dimensions (median 3), while other conditions showed median of 0 (Control: 0, Ablation-S: 0, Ablation-H: 0) despite achieving some 5-point evaluations in specific dimensions. This demonstrates that the S-OPE configuration achieves qualitatively different outcomes, not just quantitative superiority.

Discussion

Quantitative Evaluation

The quantitative evaluation results of this study provided evidence supporting the effectiveness of the CORAL framework from multiple perspectives.

The median difference of 5.7 points (30.7 vs 25, out of 35 points; Table 6) that the Test condition (S-OPE configuration) showed against the Control condition (integrated custom instruction) exceeded the minimum detectable effect size (3 points) set in power analysis, representing a practically meaningful difference. The effect size $r = 0.83$ is classified as a “very large effect” (Table 7), statistically supporting that the S-OPE configuration’s superiority is not due to chance. The score distributions visualized in Figure 6 show minimal overlap between conditions, confirming that the observed differences are not attributable to outliers but reflect consistent performance gaps.

The ablation analysis results suggest the validity of the CORAL framework’s design principles. Both the Ablation-S condition (median 27.3) with Sensemaker Cell removed and the Ablation-H condition (median 26.3) with TIDE protocol context inheritance definition removed showed significantly lower scores than the Test condition. This indicates that the essential inquiry function through strategic dialogue (Sensemaker) and the context inheritance function that

maintains semantic coherence between Cells (TIDE) each independently contribute to the CORAL framework’s effectiveness.

Integration Effect (Synergy) Analysis: From the ablation analysis data, the integration effect of components can be estimated. The difference between Ablation-S condition (TIDE present, Sensemaker absent) and Control condition is +2.3 points (27.3 - 25), which can be interpreted as the standalone effect of TIDE protocol. Similarly, the difference between Ablation-H condition (Sensemaker present, TIDE absent) and Control condition is +1.3 points (26.3 - 25), which can be interpreted as the standalone effect of Sensemaker. The simple sum of these is +3.6 points, but the actual difference between Test condition and Control condition is +5.7 points. This difference (approximately 2 points) suggests an interaction effect that occurs when Sensemaker and TIDE protocol are integrated. That is, the technical value of the CORAL framework may exceed the sum of effects of individual components. However, this integration effect estimation is an approximate calculation assuming independence between Ablation conditions, and additional experiments with a 2x2 factorial design are needed for rigorous interaction analysis.

Noteworthy is the pattern of achieving 5-point (Outstanding) evaluations (Table 8). The Test condition achieved consistently high 5-point evaluations across all dimensions (median 3), while other conditions all had a median of 0 (Control: 0, Ablation-S: 0, Ablation-H: 0). Although Ablation conditions achieved some 5-point evaluations in specific dimensions (see Figure 8), only the Test condition achieved outstanding evaluations consistently. This result indicates that the S-OPE configuration achieved the highest-level evaluation in the evaluation criteria, not just achieving higher average scores. For 5-point evaluation, the protocol’s meta-criteria require “meeting 4-point criteria plus additionally demonstrating any of strategic thinking, innovativeness, essential connection to mission, or deep insight,” suggesting that the S-OPE configuration has functionality to support these higher-order cognitive processes. The statistical significance of these findings was confirmed even after FDR adjustment for multiple comparisons (Table 9), strengthening the reliability of the results.

On the other hand, for 4-point (Excellent) evaluations, the Test condition actually had fewer (median 3.3333333 vs Control: 4, Ablation-S: 5.6666667, Ablation-H: 4.6666667). This is the flip side of the Test condition’s tendency to achieve 5 points rather than 4 points, consistent with the S-OPE configuration’s design to promote level-up from “Excellent” to “Outstanding.”

Implications for Practitioners: When aiming for high-quality outcomes in complex problem-solving tasks, the following two points are effective: (1) introduction of an essential inquiry phase through strategic dialogue, and (2) design of semantic context inheritance between processing units. These can function as design principles applicable to intellectual work utilizing AI in general, not limited to the CORAL framework.

Validity of Order Effects Control: The verification of order effects (Figure 7) confirmed that the Latin square counterbalancing implemented in this study successfully controlled for potential learning and fatigue effects. If systematic order effects existed, all conditions would exhibit similar upward (learning) or downward (fatigue) trends. The absence of such uniform

trends across conditions supports the internal validity of this study. The slight downward slope observed only in the Control condition likely reflects the inherent inconsistency of that condition—where performance depends more heavily on momentary prompt interpretation—rather than experimenter fatigue. This interpretation is consistent with the higher variability observed in the Control condition’s descriptive statistics. Conversely, the stability of the Test condition’s scores across the study period suggests that the structured approach of the S-OPE configuration provides more consistent outcomes regardless of when the task is executed.

Iterative Improvement Observed in Development Process

The development process of the S-OPE configuration (see Section) is a record of observing the iterative improvement process of the CORAL framework. In particular, the transition period from the integration experiment failure on April 28, 2025 to the emergence of the TIDE protocol on May 9, 2025 provides a concrete example of double-loop learning.

Analyzing the S-OPE development process from the perspective of Argyris & Schon’s (Argyris and Schön 1978) double-loop learning theory, a gradual evolution from single-loop learning to double-loop learning is observed.

The functional improvements in early development (January to April 27, 2025) correspond mainly to single-loop learning. The introduction of strategic dialogue functionality (April 16) and addition of orchestration functionality (April 27) were functional extensions within the existing design philosophy, remaining at improvement of problem-solving “methods.”

However, the attempt on April 28 to integrate multiple Executors into a single file and its failure became a decisive trigger forcing a review of the system’s “fundamental assumptions.” This attempt was an effort to reduce complexity by decreasing the number of custom instructions and reducing overall description volume. Behind this was an attempt to practice the tacit knowledge included in existing prompt engineering techniques: “achieve objectives with less description in custom instructions and avoid attention control breakdown.”

This integration experiment ended in failure as the LLM could no longer grasp the overall instructions, and CORAL as a whole became unstable. This was a case where integrating multiple Executors created a large single custom instruction as a result, and the descriptions attempting to control it increased descriptions of other custom instructions, ultimately failing to reduce complexity and also causing attention control breakdown. Importantly, this failure led to the discovery of the necessity of Cell division (see Section) and the importance of context inheritance.

First, if one intends to achieve thinking enhancement by structuring, verbalizing, and implementing one’s own thinking processes as AI reasoning processes, increasing description precision is necessary. This inevitably causes custom instruction description volume to increase, making custom instruction management and attention control difficult. Therefore, dividing custom instructions into functional units is necessary, and simultaneously, coordination between divided

Cells is essential. Handling “thinking” interactively with users requires sequential processing rather than MAS-like distributed processing, and processing tasks between multiple Cells while switching abstraction levels or functions needs to be done under consistent context. Hence “context inheritance” that maintains semantic coherence as task handover becomes necessary.

This can be interpreted as double-loop learning, where questioning of fundamental assumptions—“Why was that solution chosen?”—clarified the technical challenges inherent in the core concept of structuring and implementing thinking processes, leading to solutions. The rapid increase in description volume from 14,745 characters to 36,223 characters during the process leading to the TIDE protocol emergence on May 9 indicates that the emergence of TIDE established the Cell concept, solved the technical challenge of attention control breakdown due to large custom instructions, and contributed to completing the structuring and implementation of thinking processes as a practically functioning CORAL implementation.

The most important insight in this development process is the discovery that structuring and implementing thinking processes is not a pre-designed transformation process but a dynamic process that emerges through human-AI interaction. According to development records, the establishment of this design policy (May 9) occurred simultaneously with the technical implementation of the TIDE protocol, indicating that theory and implementation developed in parallel. This improvement process can be understood as the cyclical structure of “forethought phase -> performance phase -> self-reflection phase” in Zimmerman’s (Zimmerman 2002) self-regulated learning theory. Cycles of Cell design (forethought), AI execution (performance), and result observation and modification (self-reflection) were iterated, and in the process, the developer refined Cell descriptions, resulting in improved AI reasoning processes. Records show an iterative development where the developer gained new insights from AI behavior to develop Cell descriptions, and those refined Cells improved AI reasoning quality.

Integrated Theoretical Implications

Integrating the quantitative evaluation results and development process observation records, the theoretical contributions that the CORAL framework brings to human-AI collaboration become clearer.

First, the effectiveness of the S-OPE configuration demonstrated in quantitative evaluation shows that the approach of “structuring and implementing individual thinking processes” functions as a technical implementation that produces demonstrable effects, not just a theoretical construct. The fact that the Test condition showed statistically significant superiority against all comparison conditions means that the approach of structuring user cognitive characteristics and implementing them as AI reasoning processes can achieve outcomes exceeding general-purpose prompt engineering techniques (Control condition). Importantly, this comparison is not simply about which “prompt” is better. Complex strategic dialogue functions like the Sensemaker Cell (approximately 4,500 characters, including multiple thinking patterns and staged dialogue processes) technically do not function when integrated into a single prompt

due to attention control breakdown (see Section). Therefore, the difference between Test and Control conditions reflects “the value of functions made possible only through Cell division architecture,” not “ingenuity in prompt engineering.”

The ablation analysis results revealed the independent contributions of each element constituting the CORAL framework’s design principles. The performance decrease from removing the Sensemaker Cell indicates that strategic dialogue functionality contributes to artifact quality. However, whether this effect is due to “introduction of desirable difficulties” as intended by design or other mechanisms has not been verified in this study. Additionally, the performance decrease from removing the TIDE protocol’s context inheritance definition indicates that context inheritance structured across six perspectives contributes to artifact quality. The dimension-specific patterns of 5-point evaluations in Ablation conditions (Figure 8) provide additional evidence for these independent contributions: Ablation-S achieved some outstanding evaluations in dimensions related to adaptive analysis and practical solutions, while Ablation-H showed them in problem identification and impact evaluation dimensions. However, only the full S-OPE configuration achieved consistent outstanding evaluations across all dimensions, suggesting that the integration of both components is necessary for comprehensive excellence. The IQR patterns in Table 10 and the score frequency distributions visualized in Figure 9 further support this interpretation: Ablation-S showed zero variance ($IQR = 4.0-4.0$) in most dimensions (5 of 7) and Ablation-H in some (3 of 7), indicating a “ceiling effect” at 4 points, while the Test condition maintained IQRs of 4.0–5.0 across all dimensions, demonstrating consistent ability to achieve outstanding evaluations.

Considering together the double-loop learning case shown in the development process observation record (Section) and the 5-point evaluation achievement pattern shown in quantitative evaluation, the mechanism of iterative improvement promoted by the CORAL framework becomes visible. The developer used the failure of the integration experiment (April 28, 2025) as a trigger to fundamentally question existing assumptions (complexity reduction through custom instruction description volume reduction) and created a new solution called the TIDE protocol. The fact that the S-OPE configuration established as a result of this double-loop learning process became the only condition to achieve 5-point (Outstanding) evaluation can be interpreted as a concrete outcome of an improvement process based on deep reflection.

These findings suggest a new direction for personalization technology in the HCI field. While conventional adaptive systems were approaches that adjusted systems “from outside” based on user behavior patterns and preferences, the CORAL framework enables users to structure and verbalize their own thinking processes and implement them in AI systems “from within.” The model that this structuring and implementation process enables continuous improvement based on feedback and potentially promotes user metacognition in the process presents a new direction for human-AI collaboration.

Research Agenda for the HCI Community: The findings shown in this study suggest the following research directions: (1) Identification of success factors for structuring and implementing individual thinking processes—what cognitive characteristics, domain knowledge, and expression styles promote effective structuring and implementation; (2) Longitudinal

measurement of cognitive transformation through structuring and implementation—long-term changes in metacognitive ability, self-regulated learning ability, and problem-solving skills; (3) Comparison of structuring and implementation patterns across diverse user groups—verification of differences by expert/novice, language/culture, and cognitive style.

Limitations

The constraints of this study stem from bias due to proof of concept by a single developer and limitations in external validity (generalizability). While bias mitigation is attempted by having a dedicated AI agent score reports, the evaluation rubric design and operation are conducted by the developer, making complete bias elimination difficult. Human rater evaluation was not adopted because the developer serving as the rater would introduce even greater bias, undermining the validity of any such comparison. Additionally, since this is a case of structuring and implementing cognitive processes by a specific individual, the possibility that this is a special case by a special individual cannot be excluded, and verification of external validity is impossible within the scope of this study. However, for the former, verifiability is ensured by maximizing transparency, and for the latter, it is considered to be within a generally acceptable range for proof of concept (PoC).

Constraints due to language and cultural differences are also important. This study conducted the CORAL framework’s core concept of structuring and implementing individual thinking processes in Japanese, and considering the close relationship between thinking and language, this implementation may be specialized for the cognitive style of Japanese speakers. If language structure influences thinking patterns as linguistic relativity suggests, structuring and implementing thinking processes in different language families (English, Chinese, Arabic, etc.) may show different characteristics (Wolff and Holmes 2011). This language dependency poses research challenges for verifying the universality of the CORAL framework.

Technical constraints include current LLM context length limitations and response consistency, but these are expected to improve with future LLM technology development.

Regarding computational costs, it must be acknowledged that the CORAL framework is inherently a high-cost design. The S-OPE configuration uses custom instruction groups of approximately 38,000 characters, and these Cell definitions themselves consume tokens. Furthermore, Cells like the Sensemaker Cell that orient toward “dialogue” as a means of metacognition promotion inevitably consume many tokens in the dialogue process. Therefore, applications where the CORAL framework is justified are limited to areas where artifact quality has value exceeding cost, such as complex problem-solving and advanced intellectual production. The reason cost measurement was not included as an evaluation metric in this study and a high-cost reasoning model was used is that this point is self-evident and the CORAL framework is premised on pursuing artifact quality rather than efficiency. For straightforward tasks or applications where immediacy is prioritized, single custom instructions like the Control condition, or even simpler prompts, are clearly appropriate.

As a constraint of the evaluation methodology in this study, the measurement target is limited to short-term effects of final artifact (report) quality. Long-term effects of the iterative improvement cycle supported by the CORAL framework—improvement in metacognitive ability, qualitative changes in learning processes—have not been verified within the scope of this study. The current experimental design measures only artifact quality within a single session and does not capture long-term effects such as changes in learning curves, continuous improvement in problem-solving ability, and refinement of cognitive styles. This constraint suggests that longitudinal research design is needed for verification of long-term effects.

LLM Technology Evolution and Framework Validity

Although the evaluation of this study was conducted with a specific LLM, the value of the CORAL framework has structural aspects orthogonal to LLM performance improvement. This point is discussed here.

First, the structural problem of attention control is not resolved by LLM technology evolution. Even if LLM context length expands, the problem of attention breakdown for large custom instructions is not fundamentally solved. Rather, the Lost in the Middle phenomenon demonstrated by Liu et al. (Liu et al. 2024) may worsen with increased context length. Cell division in the CORAL framework (see Section) is an architecture-level solution to this structural problem and has value independent of LLM performance.

Second, the difference between Test and Control conditions shown in this study’s 4-condition comparison reflects the value of the architectural approach of “structuring and implementing individual thinking processes,” not LLM performance itself. The result that removal of both Sensemaker Cell and TIDE protocol caused performance decrease in ablation analysis (see Section) indicates the structural value provided by these components, and this value is not substituted by LLM performance improvement.

Third, the CORAL framework is based on an LLM-independent design philosophy. As stated in Section , Cells can be implemented on any platform that has the capability to explicitly invoke custom instructions. Due to this versatility, the benefits of structuring through Cell division and TIDE protocol are expected to be maintained even with more powerful models in the future.

Therefore, despite the rapid evolution of LLM technology, the value of the approach presented by the CORAL framework—“structuring and implementing individual thinking processes”—is considered to have temporal universality.

Future Directions

Future research requires reproducibility verification by other individuals and domain experts, demonstration of utility in different task domains, and comparative research on thinking process

structuring and implementation characteristics in multilingual environments. A particularly important research topic is verification of the reproducibility of the rapid improvement cycle observed in S-OPE development. In this study, development records show cross-domain theory construction and implementation refinement over a short period of several months, but whether this phenomenon is reproducible in other individuals or different contexts is unverified.

Through longitudinal research design, rigorous verification of what long-term changes continuous human-AI collaboration using the CORAL framework actually brings to user learning curves, metacognitive ability, and refinement of professional thinking processes is required. This verification is expected to clarify whether the improvement patterns observed in this study hold as general phenomena beyond individual cases.

Verification of applicability to diverse cognitive processes is also an important research direction. The S-OPE configuration of this study reflects specific cognitive styles with backgrounds in pharmaceutical research and private company practice, but elucidation of structuring and implementation process characteristics in different professional fields (creative activities, technical development, educational practice, etc.) and cognitive characteristics (analytical vs. intuitive, systematic vs. emergent, etc.) is needed.

Technical development directions include optimization of dynamic relationships between Cells, consideration of parallel processing by multiple LLMs, and implementation of real-time adaptation functions. In particular, development of technical solutions that deepen personal adaptation while reducing user cognitive load is important for improving CORAL framework practicality.

As a solution direction for the essential high cognitive load required by the CORAL framework, we propose “distributed cognitive load and collective intelligence formation through construction of an ecosystem for reusing diverse Cells specialized for individuals.” This ecosystem is expected to realize a sustainable model where the high cognitive load as initial investment is recovered as value of constructing personalized AI collaboration environments in the long term.

As theoretical development directions, formalization of iterative improvement cycles, development of effect measurement indicators for structuring and implementing cognitive processes, and construction of integrative frameworks with diverse learning theories (constructivist learning, social learning theory, etc.) are suggested. Through development of these theoretical foundations, it becomes possible to establish the proof of concept shown by the CORAL framework as a research area with broader academic and practical value.

Conclusion

This study proposed the CORAL framework as a new approach to structure and verbalize individual thinking processes and implement them as AI reasoning processes, verifying its effectiveness through proof of concept with the S-OPE configuration and rigorous quantitative evaluation. Through controlled comparative study, we demonstrated the feasibility of structuring

and implementing individual thinking processes in natural language. In complex problem-solving tasks, the S-OPE configuration showed statistically significantly higher scores compared to the Control condition (30.7 vs 25, median difference of 5.7 points, effect size $r = 0.83$, $p < 0.001$), demonstrating effectiveness.

The CORAL framework enables structured human-AI collaboration beyond the conventional “user-tool” relationship. Through statistical comparison using Mann-Whitney U test and evaluation using a 7-item evaluation rubric, the effectiveness of the CORAL framework in complex problem-solving tasks was empirically demonstrated.

Future research topics include expansion of application to diverse professional fields, verification of generalizability through large-scale multi-institutional collaborative research, and longitudinal evaluation of long-term improvement effects. Human-AI collaboration is evolving from pursuit of efficiency to amplification of individuality, and the CORAL framework is an implementable solution that concretely demonstrates one direction of this evolution. We hope that this study contributes to the research community toward realizing new forms of intellectual production that maximize individual intellectual assets through continuous improvement cycles.

Conflict of Interest Statement

The author declares no conflict of interest.

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Data Availability Statement

Zenodo DOI Reference

All data, analysis code, and research materials used in this study are available in the Zenodo repository (k-shibuki 2025).

English Translation of Key Cells (for International Readers)

For understanding promotion in international readers and the research community, English translations of all custom instructions are published in the above Zenodo repository (updated version). These translations are provided as reference materials for understanding the technical details of the CORAL framework and specific methods of structuring and implementing individual thinking processes in an international research context, provided through machine translation, and do not guarantee translation accuracy.

Statistical and Execution Environment

Statistical analysis was conducted using R version 4.5.1 (R Core Team 2025) with the following major libraries: tidyverse 2.0.0 (Wickham et al. 2019), ggplot2 3.5.2 (“Create Elegant Data Visualisations Using the Grammar of Graphics • Ggplot2” n.d.), knitr 1.50 (“Knitr - Yihui Xie” n.d.), and kableExtra 1.4.0 (Zhu et al. 2024).

Regarding the Cursor IDE version: the evaluation protocol was pre-registered and power analysis was conducted using Cursor version 0.50.4 and 0.50.5. The main evaluation trials reported in this paper were conducted using Cursor version 1.1.3 to 1.4.3. This version progression reflects the normal software update cycle during the research period and does not affect the reproducibility of the experimental design.

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